ICMA214

Advanced Calculus

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Chapter 1

The Real Numbers

1.1 Irrationality of $\sqrt{2}$

We begin by examining a proof that $\sqrt{2}$ is irrational, which despite using simple arithmetic, is surprisingly beautiful. Although constructed nearly 2000 years ago by the Pythagoreans, it still holds up today without so much as a wrinkle.

Theorem 1.1.1. *There is no rational number whose square is* 2.

Proof. A rational number is any number that can be expressed in the form p/q, where p and q are integers. Thus, what the theorem asserts is that no matter how p and q are chosen, it is never the case that $(p/q)^2 = 2$.

Assume to the contrary that there exist integers p and q satisfying

$$\left(\frac{p}{q}\right)^2 = 2.$$

We may also assume that p and q have no common factor, because if they had one, we could simply cancel it out and rewrite the fraction in lowest terms. The equation above implies

$$p^2 = 2q^2.$$

From this, we ascertain that p^2 is an even integer and hence p must be event as well (as the square of an odd number is odd). This allows us to write $p = 2r$, where r is also an integer. If we substitute $2r$ for p in equation (2), then a little algebra yields the relationship

$$2r^2 = q^2.$$

This last equation implies that q^2 is even, and hence q must also be even. Thus, we have shown that p and q are both even, and divisible by 2, which is a contradiction to our assumption that they both have no common factor. Thus, the theorem is proved. □

Fun Fact: The Pythagoreans

Pythagoras's philosophy was that music, math and nature were interchangable. Pythagoreans believed that ratios were key to understanding nature and spent much of their energy researching their properties. This led them to discover the *golden ratio* or "most beautiful number", which could be achieved by dividing a line into two so that the ratio of the small part to the large was equivalent to the ratio of the large to the whole.

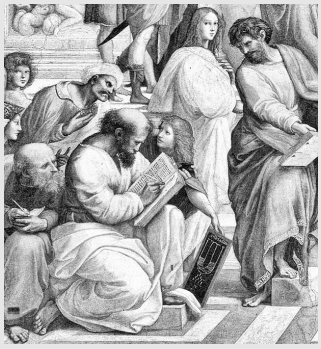


Figure 1.1: The Pythagoreans [*Credit: Wikipedia*]

For everything in the universe to be governed by ratios, everything that made sense in the universe had to be related to a proportion (i.e., rational numbers). Since Greek mathematicians studied shapes (there was no significant distinction between shapes and numbers, hence why we have things like square/triangular numbers today), it made sense to ask 'what is the ratio of a square to its diagonal'? Coincidentally enough, the diagonal of a unit square has length $\sqrt{2}$, which as we now know, is irrational. So it was probably a pretty big bummer for the Pythagoreans to have proved that $\sqrt{2}$ was irrational (they also later discovered that the golden ratio was irrational which must have been even worse).

1.3 The Axiom of Completeness

What exactly is a real number? A basic definition is that the set of real numbers $\mathbb{R}$ is an extension of the rationals $\mathbb{Q}$ in which there are no gaps. To be precise about how we phrase this assumption, we will introduce something called the *Axiom of Completeness*. However, before doing so, we have to understand bounds.

Definition 1.3.1 (Upper and Lower Bounds). *A set $A \subseteq \mathbb{R}$ is bounded above if there exists a number $b \in \mathbb{R}$ such that $a \leq b$ for all $a \in A$. The number b is called an upper bound for A. Similarly, the set A is bounded below if there exists a lower bound $l \in \mathbb{R}$ satisfying $l \leq a$ for every $a \in A$.*

Let's illustrate this fairly intuitive concept with some examples

Example 1. Let $A = (0, 5]$, i.e. $A = \{a \in \mathbb{R} \mid 0 < a \leq 5\}$, then

- 6 is an upper bound of A.

- 4 is not an upper bound of A.
- 5 is an upper bound of A.

If $A = (0,5)$, i.e. $A = \{a \in \mathbb{R} \mid 0 < a < 5\}$, then

- 6 is an upper bound of A.
- 5 is an upper bound of A, in fact it is the *least* upper bound of A.

With the notion of bounds understood, we now introduce a more specific type of bound – the *least upper bound*

Definition 1.3.2 (Least Upper Bound). *A real number s is the least upper bound for a set $A \subseteq \mathbb{R}$ if it meets the following two criteria:*

(i) s is an upper bound for A;

(ii) if b is an upper bound for A, then $s \leq b$.

The least upper bound is frequently called the *supremum* of a set A, and denoted $\sup A$. The *greatest lower bound* or *infimum* is defined in a similar way, and denoted as $\inf A$. Although a set can have multiple upper bounds, it can only have one *least* upper bound. It is fairly simple to show that a number is an upper or lower bound for a set, however, showing that a number is a supremum of infimum of a set requires a bit more work. Consider the following example

Example 2. Let $B = (0,5)$. Prove that 0 is the infimum of B.

Proof. To show that 0 is the infimum of B, we

1. For all $b \in B$, $0 \leq b$.
2. For all $c \in \mathbb{R}$, if c is a lower bound, then $c \leq 0$.

For the first condition, since $B = (0,5) = \{x \in \mathbb{R} \mid 0 < x < 5\}$, we have that for any $b \in B$, $0 \leq b$. For the second condition, let c be a lower bound of B. Then $c \leq x$ for any $x \in B$. We will show that $0 \geq c$. Assume to the contrary that $0 < c$. If $\frac{c}{2} \in B$, we are done, as we have an element in $B < c$, which violates the assumption that c is a lower bound. If $\frac{c}{2} \notin B$, we are *almost done*. Suppose we choose $c^* = \min\{\frac{c}{2}, y \in_R B\}$. We claim that $c^* \in B$. How? If $c^* = y$, since $y \in B$, we are done. If $c^* = \frac{c}{2}$, since $0 < \frac{c}{2} \leq y$, $\frac{c}{2} \in B$. We have that $c^* \in B$ and

$$c^* \leq \frac{c}{2} < c,$$

which is a contradiction to the assumption that c is a lower bound. Thus $0 \geq c$. From the 2 conditions, 0 is the greatest lower bound (infimum). □

Note however that the supremum of a set is not equivalent to the *maximum*, as the supremum of a set need not always be an element of the set itself.

Definition 1.3.4. *A real number a_0 is a maximum of a set A if a_0 is an element of A and $a_0 \geq a$ for all $a \in A$. Similarly, a number a_1 is a minimum of A if $a_1 \in A$ and $a_1 \leq a$ for every $a \in A$.*

With these definitions in mind, we can now turn our attention to the *Axiom of Completeness*, which serves as the bedrock of analysis.

Theorem (Axiom of Completeness). *Let $A \neq \emptyset$ be a subset of $\mathbb{R}$. If A has an upper bound, it has a least upper bound, called the supremum.*

We can also define the mirror version of this theorem, which we'll call the Dual Axiom of Completeness.

Theorem (Dual Completeness Axiom). *Let $B \subseteq \mathbb{R}$ and $B \neq \emptyset$. If B has a lower bound, then B has the greatest lower bound* $\inf(B)$.

1.4 Consequences of Completeness

The first application of the Axiom of Completeness is a result that works as a more natural way to mathematically express the assumption that the real line contains no gaps.

Theorem 1.4.1 (Nested Interval Property). *For each $n \in \mathbb{N}$, assume we are given a closed interval $I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$. Assume also that each I_n contains I_{n+1}. Then, the resulting sequence of closed intervals*

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \cdots$$

has a nonempty intersection. That is, $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Let $A = \{a_n \in \mathbb{R} \mid n \in \mathbb{N}\}$ and $B = \{b_n \in \mathbb{R} \mid n \in \mathbb{N}\}$. Note that $a_1 \in A$ and $b_1 \in B$. Thus $A \neq \emptyset$ and $B \neq \emptyset$. We will show that A has an upper bound and B has a lower bound. Since $I_{n+1} \subseteq I_n, \forall n \in \mathbb{N}$ and $I_n \subseteq I_1, \forall n \in \mathbb{N}$*. Thus, $a_1 \leq a_n \leq b_n \leq b_1$ for all $n \in \mathbb{N}$. We have that $a_n \leq b_1$ for all $n \in \mathbb{N}$. Thus, A has an upper bound. Similarly, we have that $a_1 \leq b_n$ for all $n \in \mathbb{N}$. Thus, B has a lower bound. By the Completeness Axiom, A has the least upper bound, say $a = \sup(A)$. By the Dual Completeness Axiom, B has the greatest lower bound $b = \inf(B)$.

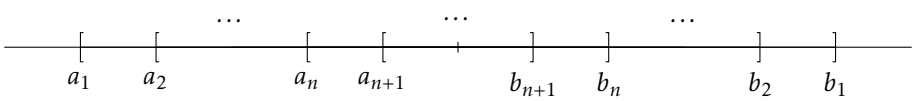


Figure 1.2: Visual representation of the nested intervals.

We will show that $a \leq b$. Assume to the contrary that $a > b$. Since $a > b$, it follows that b is *not* an upper bound of A. Therefore, there exists $n_0 \in \mathbb{N}$ such that $a_{n_0} > b$. Note that

1. $b_{n_0} \geq a_{n_0}$
2. $b_k \geq b_{n_0} \geq a_{n_0}$ for all $k \leq n_0$
3. $b_{n_0} \geq b_l \geq a_l \geq a_{n_0}$ for all $l \geq n_0$

Thus, $b_n \geq a_{n_0}$ for all $n \in \mathbb{N}$. Therefore, a_{n_0} is a lower bound of B. This is a contradiction since b is the greatest lower bound and $a_{n_0} > b$. Thus $a \leq b$.

*Because $(I_1 \supseteq I_2 \supseteq I - 3 \supseteq \cdots \supseteq I_n \supseteq I_{n+1} \supseteq \ldots)$.

We have that $a_n \leq a \leq b \leq b_n$ for all n. Thus, we have $a, b \in I_n$ for all $n \in \mathbb{N}$. Therefore, $\sum_{n=1}^{\infty} I_n \neq \emptyset$. Moreover, we have that

$$\bigcap_{n=1}^{\infty} I_n = [a, b].$$

□

As $\mathbb{Q}$ is an extension of $\mathbb{N}$ and $\mathbb{R}$ is in turn an extension of $\mathbb{Q}$, we will next show how $\mathbb{Q}$ and $\mathbb{N}$ are subsets of $\mathbb{R}$.

Theorem 1.4.2 (Archimedean Property).

(i) For any $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.

(ii) For any $y \in \mathbb{R}$, if $y > 0$, then there exists $n \in \mathbb{N}$ satisfying $0 < \frac{1}{n} < y$

Proof. Part (i) of the proposition essentially states that $\mathbb{N}$ is not bounded above. Assume to the contrary that $\mathbb{N}$ is bounded above. Since $\mathbb{N} \neq \emptyset$ and $\mathbb{N}$ is bounded, the Axiom of Completeness implies that $\mathbb{N}$ has the least upper bound, say $y_0 \in \mathbb{R}$. Thus, $y_0 - 1$ is not an upper bound of $\mathbb{N}$. We have that there is $n_0 \in \mathbb{N}$ such that

$$y_0 - 1 < n_0 \leq y_0$$

And hence

$$(y_0 - 1) + 1 < n_0 + 1$$

Thus, $y_0 < n_0 + 1$ and $n_0 + 1 \in \mathbb{N}$. This is a contradiction, as we defined y_0 to be the supremum of $\mathbb{N}$. For part (ii), we apply (i) for $x = \frac{1}{y} > 0$. Let $y > 0$. By (1), there exists $n > x = \frac{1}{y} > 0$. Therefore $0 < \frac{1}{n} < y$ □

Using the Archimedean Property, we can show the density of $\mathbb{Q}$ in $\mathbb{R}$.

Theorem 1.4.3 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *For every two real numbers a and b with $a < b$, there exists a rational number r satisfying $a < r < b$.*

Proof. As r is a rational number, this is equivalent to saying that there exists $m \in \mathbb{Z}$ and $n \in \mathbb{N}$ satisfying

$$a < \frac{m}{n} < b.$$

We'd like to choose n large enough such that consecutive increments are too small to 'step over' the interval (a, b). By the Archimedean Property, we can choose $n \in \mathbb{N}$ large enough such that

$$\frac{1}{n} < b - a.$$

Observe now that

$$a < \frac{m}{n} < b \iff na < m < nb,$$

so we turn our attention to choosing m such that it is the smallest integer greater than na. Choose $m \in \mathbb{Z}$ such that

$$m - 1 \leq na < m.$$

This immediately yields $a < m/n$ but for the other half of the inequality, we have

$$\begin{aligned} m &\leq na + 1 \\ &< n\left(b - \frac{1}{n}\right) + 1 \\ &= nb. \end{aligned}$$

As $m < nb$ implies $m/n < b$, we have

$$a < m/n < b,$$

as desired. □

Existence of Square Roots

We have previously proved that $\sqrt{2}$ is not a rational number. However, we never proved that it even... exists. Can we do so?

Theorem 1.4.5. *There exists a real number $\alpha_0 > 0$ satisfying $a_0^2 = 2$.*

Proof. Let $T = \{\alpha \in \mathbb{R} \mid \alpha^2 < 2\}$. To apply the Axiom of Completeness, we require T to be non-empty and bounded above. To verify that T is non-empty, observe that

$$1^2 = 1 < 2.$$

Thus, $1 \in T$ and $T \neq \emptyset$. For the second requirement, we will show that 2 is an upper bound of T. Let $\alpha \in T$. If $\alpha < 0$, then we are done, as $\alpha < 2$. For the case that $\alpha \geq 0$, assume to the contrary that $\alpha \geq 2$. Then

$$2\alpha \leq \alpha \cdot \alpha = \alpha^2 < 2.$$

Thus, we have that

$$2 \cdot 2 \leq \alpha \cdot 2 < 2,$$

which is a contradiction. We conclude that 2 is an upper bound of T. By the Axiom of Completeness, T has the supremum, say $\sup T = \alpha_0$[†]. We have that $\alpha_0^2 < 2$, $\alpha_0^2 = 2$ or $\alpha_0^2 > 2$. We will show the impossibilities of $\alpha_0^2 < 2$ and $\alpha_0^2 > 2$. Assume to the contrary that $\alpha_0^2 < 2$. We claim that there exists $n_0 \in \mathbb{N}$ such that

$$\left(\alpha_0 + \frac{1}{n_0}\right)^2 < 2.$$

We want to show that

$$\left(\alpha_0 + \frac{1}{n_0}\right)^2 = \alpha_0^2 + 2\alpha_0 \cdot \frac{1}{n_0} + \frac{1}{n_0^2} < 2.$$

To do so, we can simplify the above as follows

$$\begin{aligned} \frac{2\alpha_0 n_0 + 1}{n_0^2} &< \underbrace{2 - \alpha_0^2}_{\text{positive}} \\ \frac{2\alpha_0 n_0}{n_0^2} < \frac{2\alpha_0 n_0 + 1}{n_0^2} &< 2 - \alpha_0^2 \\ 0 < \frac{2\alpha_0}{n_0} &< 2 - \alpha_0^2. \end{aligned} \tag{$*$}$$

[†]This does not automatically imply that $\alpha_0^2 = 2$.

Consider $\frac{n_0}{2-\alpha_0^2} \cdot (*)$

$$0 < \frac{2\alpha_0}{2\alpha_0^2} < n_0.$$

By the Archimedean Property, there exists $n_0 \in \mathbb{N}$ satisfying this inequality. Thus, $\alpha_0 + \frac{1}{n_0} \in T$, which contradicts the property of $\alpha_0 = \sup T$. Therefore, $\alpha_0^2 \geq 2$. Verifying the mirror case that $\alpha_0^2 > 2$ is impossible can be done similarly, by instead assuming to the contrary that $\alpha_0^2 > 2$ and considering

$$\left(\alpha_0 - \frac{1}{n_1}\right)^2 > 2.$$

Therefore, α_0^2 must be equal to two, concluding the proof. □

1.5 Cardinality (Extra)

Now we kind of have an image of the real numbers as consisting of rational and irrational numbers. There is a temptation to think of the two as being mixed together in equal proportions, however, this is not the case. As Georg Cantor showed precisely, the irrationals far outnumber the rationals in making up $\mathbb{R}$.

1-1 Correspondence

The term *cardinality* is used to refer to the size of a set. Although we can use cardinalities to compare finite sets, we can also do so by attempting to put sets into a 1-1 correspondence with each other. The advantage of this method is that it works on sets that are infinite.

Definition 1.5.1. *A function $f : A \to B$ is one-to-one (1-1) if $a_1 \neq a_2$ in A implies that $f(a_1) \neq f(a_2)$ in B. The function f is onto if, given any $b \in B$, it is possible to find an element $a \in A$ for which $f(a) = b$.*

A function that is both 1-1 and onto provides us with exactly what we mean when we say a 1-1 correspondence between two sets. Being 1-1 means that there are no two elements of A that correspond to B, and the property of being onto ensures each element of B corresponds to something in A.

Definition 1.5.2. *The set A has the same cardinality as B if there exists $f : A \to B$ that is 1-1 and onto. In this case, we write $A \sim B$.*

Countable Sets

Now that we understand the notion of cardinality and 1-1 correspondence, we can introduce countable sets.

Definition 1.5.5. *A set A is countable if $\mathbb{N} \sim A$. An infinite set that is not countable is called an uncountable set.*

Examples of countable sets include $\mathbb{Z}$ or the set of even natural numbers $E = \{2, 4, 6, \dots\}$. Why? For the former, it's rather simple. We want to create a mapping that is 1-1 and ensures every

element of $\mathbb{Z}$ gets "hit" by something in $\mathbb{N}$. We can do so with the mapping

$$\begin{array}{cccccccc} \mathbb{N} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots \\ & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\ \mathbb{Z} & 0 & -1 & 1 & -2 & 2 & -3 & 3 & \dots \end{array}$$

For the set of even numbers, we can just use the mapping

$$\begin{array}{ccccccc} \mathbb{N} & 1 & 2 & 3 & 4 & \dots & n & \dots \\ & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \dots & \updownarrow & \dots \\ E & 2 & 4 & 6 & 8 & \dots & 2n & \dots \end{array}$$

It might be tempting to think that all infinite sets are countable. However, it turns out that $\mathbb{R}$ is *uncountable*. The real numbers outnumber the natural numbers in such a way that it is not possible to map $\mathbb{N}$ onto $\mathbb{R}$ (the set $\mathbb{Q}$ on the other hand is countable).

Chapter 2

Sequences and Series

2.2 The Limit of a Sequence

An understanding of infinite series depends heavily on a clear understanding of regular sequences – in fact, most concepts in analysis can be reduced to statements about the behavior of sequences. So, we will begin by investigating sequences.

Definition 2.2.1 (Sequence). *$a : \mathbb{N} \to \mathbb{R}$ is called a sequence of real numbers, where $a(n) = a_n$.*

Consider the following sequence

$$1, 11, 21, 1211, 111221, \ldots$$

We see that this sequence is merely the frequency of the contiguous numbers. If we continue this sequence, will we eventually see 4? How about 5? In most cases, what happens at the beginning of a sequence is of little importance. We are more generally concerned with the behavior of the 'tail' of a given sequence, or rather what it *converges* to.

Definition 2.2.3 (Convergence). *Let (a_n) be a real sequence. (a_n) converges to $L \in \mathbb{R}$ if for each $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that for every $n \geq N$, we have that*

$$|a_n - L| < \varepsilon.$$

Note that if (a_n) converges to L, then we write

$$\lim_n a_n = L.$$

As a warmup, let's use this definition to prove that some sequences converge to a value.

Example 3. We will show that $(1) = (1, 1, 1, \ldots)$ converges to $(a_n = 1), \forall n \in \mathbb{N}$.

Let $\varepsilon > 0$. We want to choose N such that for any $n \geq N$,

$$\begin{aligned} |a_n - L| &= |1 - 1| \\ &= |0| \\ &= 0 \\ &< \varepsilon. \end{aligned}$$

In this instance, we are free to choose N as we wish. Suppose $N = 1$. Thus, for each $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - 1| < \varepsilon$.

We can also do a less trivial (although still fairly clear) example.

Example 4. Let's prove that $\left(\frac{1}{n}\right)$ converges to 0.

Let $\varepsilon > 0$. We want to choose N such that for any $n \geq N$,

$$\left|\frac{1}{n} - 0\right| < \varepsilon \iff 0 < \frac{1}{n} < \varepsilon.$$

By the Archimedean Property, there is $N \in \mathbb{N}$ such that

$$0 < \frac{1}{N} < \varepsilon.$$

Therefore, for every integer $n \geq N$,

$$0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

We have that for all $n \geq N$,

$$\begin{aligned} |a_n - 0| &= \left|\frac{1}{n} - 0\right| \\ &= \left|\frac{1}{n}\right| \\ &= \frac{1}{n} \\ &< \varepsilon. \end{aligned}$$

Thus, $\lim_n \frac{1}{n} = 0$.

Quantifiers

The logic involved for convergence proofs is quite complicated and is quite dependent on the use of quantifiers like "for all" and "there exists". The basic template for a proof that the sequence (x_n) converges to x is as follows:

1. Let $\epsilon > 0$ be arbitrary.
2. Demonstrate a choice for $N \in \mathbb{N}$ (usually requires the most work and done in scratch).
3. Show that N actually works.
4. Assume $n \geq N$.

5. With N well chosen, derive the inequality $|x_n - x| < \epsilon$.

Furthermore, we should also note the uniqueness of limits.

Theorem 2.2.7 (Uniqueness of Limits). *The limit of a sequence, when it exists, is unique.*

Proof. Let (a_n) be a sequence and $L_1 \in \mathbb{R}$. Assume that $\lim_n a_n = L_1$. Let $L_2 \in \mathbb{R}$ be a value such that $\lim_n a_n = L_2$. We will show that they are the same. To do so directly, we want to show $\forall \varepsilon > 0, 0 \leq a < \varepsilon \implies a = 0$. Let $\varepsilon > 0$. Choose $\varepsilon' = \frac{\varepsilon}{2} > 0$. Thus, there are $N_1, N_2 \in \mathbb{N}$ such that if $n_1 \geq N_1$, then $|a_n - L_1| < \varepsilon'$ and if $n_2 \geq N_2$, then $|a_n - L_2| < \varepsilon'$.

Choose $N = \max\{N_1, N_2\}$. Thus, if $n \geq N$, then $|a_n - L_1| < \varepsilon'$ and $|a_n - L_2| < \varepsilon'$. We have that

$$\begin{aligned}
|L_1 - L_2| &= |L_1 - a_n + a_n - L_2| \\
&\leq |L_1 - a_n| + |a_n - L_2| \\
&< \varepsilon' + \varepsilon' \\
&= 2\varepsilon' \\
&= 2\left(\frac{\varepsilon}{2}\right) \\
&= \varepsilon
\end{aligned}$$

Thus, for any $\varepsilon > 0$, we have $0 \leq |L_1 - L_2| < \varepsilon$. We can conclude that $L_1 = L_2$. □

Divergence

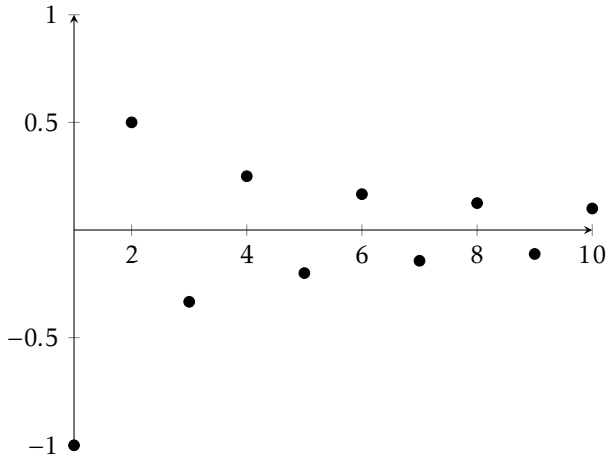
It is also possible for a sequence to not converge to any value, in which case we say it *diverges*. More formally:

Definition 2.2.9 (Divergence). *If (a_n) does not converge to any $L \in \mathbb{R}$, we say that (a_n)* ***diverges****.*

We can show how to prove that a series diverges with the following example.

Example 5. 1. Show that (a_n) converges where $a_n = \frac{(-1)^n}{n}$ for $n \in \mathbb{N}$.

Proof. As we don't know L, we can try drawing a picture to see the trend of the sequence.



We claim that $L = 0$. We will show that $\lim_n a_n = 0$. Let $\varepsilon > 0$. We want to choose $N \in \mathbb{N}$ such that for all $n \geq N$

$$\begin{aligned} |a_n - L| &< \varepsilon \\ \left|\frac{(-1)^n}{n} - 0\right| &< \varepsilon \\ \left|\frac{(-1)^n}{n}\right| &< \varepsilon. \end{aligned}$$

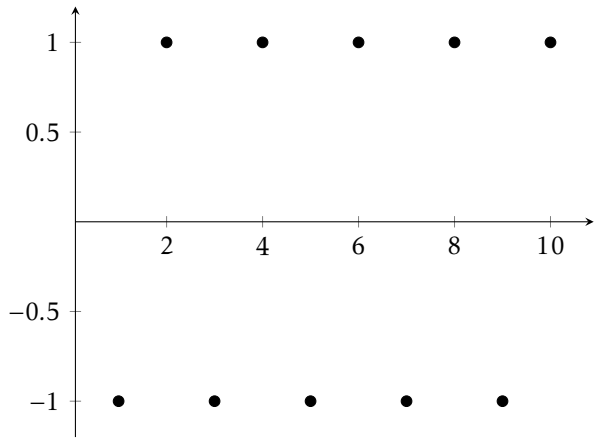
If N has the property $\left|\frac{1}{N}\right| < \varepsilon$, then $\left|\frac{1}{n}\right| < \varepsilon$. Can N has this property? Yes, by the Archimedean Property. For any $n \in \mathbb{N}$, if $n \geq N$, then $\frac{1}{n} \leq \frac{1}{N}$. Thus, for any $n \in \mathbb{N}$, we have

$$\begin{aligned} |a_n - L| &= \left|\frac{(-1)^n}{n} - 0\right| \\ &= \frac{1}{n} \\ &\leq \frac{1}{N} \\ &< \varepsilon. \end{aligned}$$

Thus, $\lim_n \frac{(-1)^n}{n} = 0$. □

2. Is $((-1)^n)$ convergent?

Again, we can try plotting.



Proof. We will show that $(a_n) = ((-1)^n)$ does not converge to any $L \in \mathbb{R}$. Assume to the contrary that there exists $L \in \mathbb{N}$ such that $\lim_n (-1)^n = L$. Let $\varepsilon > 0$. There is $N \in \mathbb{N}$ such that if $n \geq N$

$$|(-1)^n - L| < \varepsilon.$$

If n is even, then

$$|(-1)^n - L| = |1 - L| < \varepsilon. \tag{$*$}$$

If n is odd, then

$$|(-1)^n - L| = |-1 - L| < \varepsilon. \tag{$**$}$$

Let $\varepsilon_0 = 1 > 0$. Since $\lim_n (-1)^n = L$, there is $N \in \mathbb{N}$ such that if $n \geq N$, then $|(-1)^n - L| < \varepsilon_0 = 1$.

- *Case 1 (n is even)*

$$|(-1)^n - L| = |1 - L| < \varepsilon_0 = 1. \tag{B}$$

- *Case 2 (n is odd)*

$$|(-1)^n - L| = |-1 - L| = |1 + L| < \varepsilon_0 = 1. \tag{B}$$

Using the triangle inequality, we can add the two cases

$$\begin{aligned} |(1 + L) + (1 - L)| &\leq |1 + L| + |1 - L| \\ &< 1 + 1 \\ &= 2. \end{aligned}$$

Thus, $2 = |2| < 2$. Of course, this is a contradiction. Therefore, $(-1)^n$ diverges. □

2.3 The Algebraic and Order Limit Theorems

By constructing a rigorous definition for convergence, we can confidently prove statements about convergent sequences in general. Ultimately, the goal is not computation, but rather to determine the behavior of limits with respect to mathematical manipulations we perform on them.

As an example to begin with, we will prove that convergent sequences are bounded.

Definition 1 (Boundedness). *A sequence (a_n) is bounded if there is $M > 0$ such that for all $n \in \mathbb{N}$, we have*

$$|a_n| < M.$$

Geometrically, this means we can find an interval $[-M, M]$ that contains every term in the sequence (a_n).

Theorem 2.3.2. *A convergence sequence is bounded.*

Proof. Assume that a sequence (a_n) converges to $L \in \mathbb{R}$. Let $\varepsilon = 1 > 0$. We want to find $M \in \mathbb{R}$ such that $|a_n| < M$ for all $n \in \mathbb{N}$. Since $\varepsilon = 1 > 0$, there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon = 1$ for all $n \geq N$. We have that $||a_n| - |L|| < |a_n - L| < \varepsilon = 1$ for all $n \geq N$. Thus, $|a_n| - |L| < ||a_n| - |L|| < 1$ for all $n \geq N$. Therefore, $|a_n| < |L| + 1$ for all $n \geq N$. Let

$$N = \max\{|L| + 1, |a_1, |a_2|, \ldots, |a_{N-1}|\}.$$

Thus $|a_n| \leq M$ for all $n \in \mathbb{N}$. □

The following theorems will now show that sequences behave extremely well with respect to many basic mathematical operations.

Theorem 2.3.3 (Algebraic Limit Theorem). *For $\lim_n a_n = L$ and $\lim_n b_n = M$, for $L, M \in \mathbb{R}$, we have the following properties*

(i) $\lim_n c \cdot a_n = c \cdot \lim_n a_n$

(ii) $\lim_n (a_n + b_n) = \lim_n a_n + \lim_n b_n$

(iii) $\lim_n (a_n \cdot b_n) = \lim_n a_n \cdot \lim_n b_n$

(iv) $\lim_n \left(\frac{a_n}{b_n}\right) = \frac{\lim_n a_n}{\lim_n b_n}$, *where for all $n \in \mathbb{N}$, $b_n \neq 0, M \neq 0$.*

Proof.

(i) Let $\varepsilon > 0$. Since $\lim_n a_n = L$, there is $N \in \mathbb{N}$ such that if $n \geq N$, then $|a_n - L| < \frac{\varepsilon}{c}$. One small problem with this: c could be zero.

Let $c \in \mathbb{R}$. In the case that $c = 0$, then $ca_n = 0$ and $cL = 0$. Thus $\lim_n ca_n = \lim_n 0 = 0$. Let us now examine the cases where $c \neq 0$. Let $\varepsilon_c = \frac{\varepsilon}{|c|} > 0$. Since $\lim_n a_n = L$, there is $N \in \mathbb{N}$ such that if $n \geq N$, then $|a_n - L| < \varepsilon_c = \frac{\varepsilon}{|c|}$. Thus, if $n \geq N$, then $|ca_n - cL| < |c|\varepsilon = \varepsilon$.

We can conclude that $\lim_n ca_n = cL = c \cdot \lim_n a_n$.

(ii) Let $\varepsilon > 0$. Since $\lim_n a_n = L$ and $\lim_n b_n = M$, there exist $N_1, N_2 \in \mathbb{N}$ such that if $n_1 \geq N_1$, then $|a_{n_1} - L| < \frac{\varepsilon}{2}$ and if $n_2 \geq N_2$, then $|b_{n_2} - M| < \frac{\varepsilon}{2}$. Choose $N = \max\{N_1, N_2\}$ to guarantee both equalities hold. Let $n \in \mathbb{N}$ such that $n \geq N$.

$$\begin{aligned}
|(a_n + b_n) - (L + M)| &= |(a_n - L) + (b_n - M)| \\
&\leq |a_n - L| + |b_n - M| && \text{(Triangle inequality)} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon
\end{aligned}$$

Thus $\lim_n (a_n + b_n) = L + M = \lim_n a_n + \lim_n b_n$.

(iii) Let $\varepsilon > 0$. Since $\lim_n a_n = L$ and $\lim_n b_n = M$, there exist $N_1, N_2 \in \mathbb{N}$ such that if $n_1 \geq N_1$, then $|a_{n_1} - L| < \frac{\varepsilon}{2}$ and if $n_2 \geq N_2$, then $|b_{n_2} - M| < \frac{\varepsilon}{2}$. Choose $N = \max\{N_1, N_2\}$ to guarantee both equalities hold. Let $N = \max\{N_1, N_2\}$ to guarantee both equalities hold. Let $n \in \mathbb{N}$ such that $n \geq N$.

$$\begin{aligned}
|a_n b_n - LM| &= |a_n b_n - a_n M + a_n M + -LM| \\
&= |a_n b_n - a_n M| + |a_n M - LM| \\
&= |a_n| \underbrace{|b_n - M|}_{<\varepsilon} + \underbrace{|a_n - L|}_{<\varepsilon} |M|
\end{aligned}$$

Since (a_n) is convergent, it is bounded, say, $0 \leq |a_n| < A$, for all $n \in \mathbb{N}$. Choose $N^* \geq N$

$$|b_n - M| < \frac{\varepsilon}{2A}, \quad |a_n - L| < \frac{\varepsilon}{2(|M| + 1)}.$$

Let $n \in \mathbb{N}$ and $n \geq N^*$. For our inequality above

$$\begin{aligned}
|a_n b_n - LM| &\leq |a_n||b_n - M| + |a_n - L||M| \\
&< A \cdot \frac{\varepsilon}{2A} + \frac{\varepsilon}{2(|M| + 1)} \cdot M \\
&= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \cdot \underbrace{\frac{M}{(|M| + 1)}}_{<1} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon
\end{aligned}$$

(iv) It suffices to prove that baby quotient rule, i.e.,

$$\lim_n \frac{1}{b_n} = \frac{1}{M}.$$

Let $\varepsilon > 0$. Since $\lim_n b_n = M$, there is $N \in \mathbb{N}$ such that if $n \geq N$, then

$$|b_n - M| < \varepsilon.$$

Since (b_n) is convergent, there is $B > 0$ such that for all $n \in \mathbb{N}$, $|b_n| < B$. Let $n \geq N$, then

$$\left| \frac{1}{b_n} - \frac{1}{M} \right| = \left| \frac{M - b_n}{b_n M} \right|$$

Let $\varepsilon_1 = \min\{\varepsilon, \frac{\varepsilon|M|^2}{2}\} > 0$. Since $\lim_n b_n = M \neq 0$, there is $N_1 \in \mathbb{N}$ such that if $n \geq N_1$, then $|b_n - M| < \varepsilon_1$. Thus,

$$\begin{aligned}\left|\frac{1}{b_n} - \frac{1}{M}\right| &= \left|\frac{M - b_n}{b_n M}\right| \\ &< \frac{\varepsilon_1}{|b_n||M|} \\ &< \frac{\frac{|M|^2}{2}}{\frac{|M|}{2} \cdot |M|} \\ &= \varepsilon.\end{aligned}$$

□

Limits and Order

Limits can also be computed from each individual component sequence provided that each component limit exists.

Theorem 2.3.4. *Order Limit Theorem Let (a_n) and (b_n) be such that $\lim_n a_n = a$ and $\lim_n b_n = b$.*

1. *If for all $n \in \mathbb{N}, a_n \geq 0$, then $a \geq 0$.*
2. *If for all $n \in \mathbb{N}, a_n \geq b_n$, then $a \geq b$.*

Proof.

1. Assume that for all $n \in \mathbb{N}, a_n \geq 0$. Assume to the contrary that $\lim_n a_n = a < 0$. Let $\varepsilon = |a| = -a > 0$. Thus, there is $N \in \mathbb{N}$ such that for all $n \geq N$, we have

$$a_n - a \leq |a_n - a| < \varepsilon = -a \iff a_n < -a + a = 0.$$

This is a contradiction. Thus, $a \geq 0$.

2. Assume that for all $n \in \mathbb{N}$ that $a_n \geq b_n$. Define a sequence (c_n) by

$$c_n = a_n - b_n \geq 0.$$

Then we have

$$\begin{aligned}\lim_n c_n &= \lim_n (a_n - b_n) \\ &= \lim_n a_n - \lim_n b_n \\ &= a - b.\end{aligned}$$

Thus c_n converges. From (1), since $c_n \geq 0$ for all n, we have $\lim_n c_n = a - b \geq 0$.

□

2.4 The Monotone Convergence Theorem

We showed so far that convergent sequences are bounded. The converse is definitely not true. However, if the sequence is bounded *and* monotone, then it in fact does converge. First let's clear up what it means for a sequence to be monotone.

Definition 2.4.1. *A sequence (a_n) is increasing if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$ and decreasing if $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$. A sequence is monotone if it is either increasing or decreasing.*

For example, $(a_n) = \frac{n+1}{n}$ is decreasing, $(a_n) = \left(\frac{(-1)^n}{n}\right)$ is neither increasing or decreasing, and $a_n = (0) = (0, 0, 0, \dots)$ is both increasing and decreasing.

Theorem 2.4.2 (Monotone Convergence Theorem). *If (a_n) is monotone and bounded, then (a_n) is a convergent sequence.*

Proof. Without loss of generality, we assume that (a_n) is increasing and bounded. That means for all $m, n \in \mathbb{N}$, we have

1. $a_m \leq a_n$ if $m \leq n$.
2. $|a_n| \leq M, \exists M \in \mathbb{R}$.

Let $A = \{a_n \mid n \in \mathbb{N}\}$. Note that $A \neq \emptyset$ and A is bounded above. Therefore, by the AoC, A has the supremum, say $\sup A$. We claim that

$$\lim_n a_n = \sup A.$$

Let $\varepsilon > 0$ and denote $\sup A = a$. We know that $a - \varepsilon$ is not an upper bound. Thus, there is an element $a_N \in A$ such that

$$a_N > a - \varepsilon.$$

Since (a_n) is increasing, for all $n \geq N$, we have $a_n \geq a_N$ and

$$a - \varepsilon < a_N \leq a_n < a.$$

Let us rewrite this in the formal definition.

$$\begin{aligned} a - \varepsilon < a_n \leq a_n \\ -\varepsilon < a_n - a \leq 0 < \varepsilon \\ |a_n - a| < \varepsilon \end{aligned}$$

Therefore, for $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$, if $n \geq N$, then $|a_n - a| < \varepsilon$. We can conclude that $\lim_n a_n = a$. □

The Monotone Convergence Theorem is extremely useful for the studenet of infinite series, as it asserts the convergence of a sequence without explicit mention of the actual limit.

2.5 Subsequences and the Bolzano-Weierstrass Theorem

Next, we develop the concept of subsequences.

Definition 2.5.1 (Subsequences). *Let (a_n) be a sequence. Let $n : \mathbb{N} \to \mathbb{N}$ be a strictly increasing sequence/function, i.e., $n(k) < n(l)$ for all $k, l \in \mathbb{N}$ such that $k < l$. (a_{n_k}) is called a subsequence of (a_n).*

The Bolzano-Weierstrass Theorem

For bounded sequences, it turns out that is is always possible to find at least one convergent subsequence.

Theorem 2.5.5 (Bolzano-Weierstrass Theorem). *Every bounded sequence contains a convergent subsequence.*

Proof. Let (a_n) be a bounded sequence. Thus, there is $M \in \mathbb{R}$ such that

$$|a_n| \leq M.$$

Define $I_0 := [x_0, y_0] = [-M, M]$. For each $k \in \mathbb{N}$, consider

$$\begin{aligned} I_{k-1} &= [x_{k-1}, y_{k-1}] \\ &= \left[x_{k-1}, \frac{x_{k-1}+y_{k-1}}{2}\right] \cup \left[\frac{x_{k-1}+y_{k-1}}{2}, y_{k-1}\right] \\ &= J_{k_1} \cup J_{k_2}. \end{aligned}$$

Choose $i_o \in \{1, 2\}$ such that $J_{k_{i_o}}$ contains infinitely many terms of a_n's. Define

$$\begin{aligned} I_{k+1} &= [x_{k+1}, y_{k+1}] \\ &= J_{k_{i_o}} \\ &\subseteq I_k. \end{aligned}$$

Thus, we have that the closed interval I_n's satisfy

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$$

and I_k contains infinitely many terms of a_n's for each $k \in \mathbb{N}$. Note that $|I_k| = \frac{2M}{2^k}$ for $k \in \mathbb{N}$ and $\lim_k |I_k| = 0$. Since $I_1 \supseteq I_2 \supseteq \dots \supseteq I_n \supseteq \dots$ and I_n is closed and $I_n \neq \emptyset$ for all $n \in \mathbb{N}$, the Nested Interval Property implies that

$$\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset.$$

Let $a \in \bigcap_{n \in \mathbb{N}} I_n$. Since $I_1 \neq \emptyset$ and contains infinitely many a_n's, there is $a_{n_1} \in I_1$. For n_2, since $I_2 \neq \emptyset$ and contains infinitely many a_n's, there is $n_2 > n_1$ such $a_{n_2} \in I_2$. For each $k \in \mathbb{N}$, we can choose $a_{n_{k+1}} \in I_{k+1}$ such that $n_{k+1} > n_k$. Thus, (a_{n_k}) is a subsequence of (a_n). Since $|I_n| = \frac{2M}{2^n}$,

$$\lim_k a_{n_k} = a.$$

□

2.6 The Cauchy Criterion

Cauchy Sequences allow us to use a more relaxed method of proving convergence, so that we do not have to come up with L itself.

Definition 2.6.1 (Cauchy Sequences). *(a_n) is called a Cauchy sequence if for all $\varepsilon > 0$, there*

is $N \in \mathbb{N}$, such that for all $m, n \in \mathbb{N}$, if $m, n \geq N$, then

$$|a_m - a_n| < \varepsilon.$$

Are convergent sequences and Cauchy sequences the same? Thankfully, they are.

Theorem 2.6.2. *Every convergent sequence is a Cauchy sequence.*

We will show the proof for this later below. Since we have just learned about the Bolzano-Weierstrass Theorem, it is also useful to know the next lemma.

Lemma 2.6.3. *Cauchy sequences are bounded.*

Before continuing on to the next most important result, let's demonstrate how to show a sequence is Cauchy.

Example 6. 1. Show $(a_n) = (1)$ is Cauchy.

Note that for any $m, n \in \mathbb{N}$

$$|a_m - a_n| = |1 - 1| = 0.$$

Let $\varepsilon > 0$. Choose $N = 1 \in \mathbb{N}$. For any $m, n \in \mathbb{N}$ such that $m, n \geq N = 1$, we have

$$|a_m - a_n| = |1 - 1| = 0 < \varepsilon.$$

Thus, (a_n) is a Cauchy Sequence.

2. Consider (s_n), where

$$s_n = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}, \quad n \in \mathbb{N}.$$

We will show that (s_n) is a Cauchy Sequence. Let $\varepsilon > 0$. WLOG assume $n \geq m$. Observe that for any $m, n \in \mathbb{N}$,

$$\begin{aligned}
|s_n - s_m| &= \left| \frac{1}{2^{m+1}} + \frac{1}{2^{m+2}} + \cdots + \frac{1}{2^n} \right| \\
&= \frac{1}{2^m} \underbrace{\left| \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^{n-m}} \right|}_{<1} \\
&< \frac{1}{2^m} \cdot 1 \\
&= \frac{1}{2^m} \\
&< \frac{1}{m}.
\end{aligned}$$

The last equality comes from the fact that $m < 2^m, \forall n \in \mathbb{N}$. This can be proven by induction. Continuing on, we choose $N \in \mathbb{N}$ such that $0 < \frac{1}{N} < \varepsilon$, which by the Archimedean property,

is totally doable. Thus, for any $m, n \in \mathbb{N}$ such that $m, n \geq N$, we have

$$\begin{aligned} |s_n - s_m| &< \frac{1}{m} \\ &\leq \frac{1}{N} \\ &< \varepsilon. \end{aligned}$$

Thus, (s_n) is a Cauchy Sequence.

3. Let

$$s_n = \frac{1}{0!} + \frac{1}{1!} + \cdots + \frac{1}{(n-1)!}.$$

We will prove that this is a Cauchy sequence. Let $\varepsilon > 0$. Again, WLOG, assume $m, n \in \mathbb{N}$ with $n \geq m$. We have

$$\begin{aligned} |s_n - s_m| &= \left| \frac{1}{m!} + \frac{1}{(m+1)!} + \cdots + \frac{1}{(n-1)!} \right| \\ &= \frac{1}{(m-1)!} \left| \frac{1}{m} + \frac{1}{m(m+1)} + \cdots + \frac{1}{m(m+1)\ldots(m+(n-m))} \right| \\ &\leq \underbrace{\frac{1}{(m-1)!}}_{<\frac{1}{m-1}} \underbrace{\left| \frac{1}{m} + \frac{1}{m^2} + \frac{1}{m^3} + \cdots + \frac{1}{m^{n-m}} \right|}_{<1} \\ &< \frac{1}{m-1}. \end{aligned}$$

Choose $N \in \mathbb{N}$ such that $0 < \frac{1}{N-1} < \varepsilon$. For any $m, n \in \mathbb{N}$ such that $m, n \geq N$, we have

$$\begin{aligned} |s_n - s_m| &< \left| \frac{1}{m-1} \right| \\ &< \frac{1}{N-1} \\ &< \varepsilon. \end{aligned}$$

Thus, (s_n) is a Cauchy sequence.

Finally, we will prove the Cauchy Criterion

Theorem A. *sequence (a_n) is convergent if and only if it is a Cauchy sequence.*

Proof.

$(\Rightarrow)$

Let (a_n) be a convergent sequence. Thus, there is $L \in \mathbb{R}$ such that

$$\lim_n a_n = L.$$

We will show that (a_n) is a Cauchy sequence. Let $\varepsilon > 0$. Since $\lim_n a_n = L$, there is $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$, if $n \geq N$, then

$$|a_n - L| < \frac{\varepsilon}{2}.$$

Observe that for $n \geq m \geq N$, we have

$$\begin{aligned}
|a_n - a_m| &= |a_n - L + L - a_m| \\
&\leq \underbrace{|a_n - L|}_{<\varepsilon/2} + \underbrace{|L - a_m|}_{<\varepsilon/2} && \text{(Triangle ineq.)} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}$$

Thus, for any $m, n \in \mathbb{N}$, such that $m, n \geq N$, we have $|a_n - a_m| < \varepsilon$. We can conclude that (a_n) is a Cauchy sequence.

($\Leftarrow$)

Assume that (a_n) is a Cauchy sequence. We want to show that (a_n) is convergent. To do so, we have to show that

1. (a_n) is a bounded sequence.
2. (a_n) has a convergent subsequence (which converges to L). We can do so using the Bolsano-Weierstrass Theorem.
3. L is the limit of (a_n)

1. Let $\varepsilon_1 = 1 > 0$. By the property of Cauchy sequences, there is $N_1 \in \mathbb{N}$ such that for any $m, n \in \mathbb{N}$, if $m, n \geq N_1$, then
$$|a_m - a_n| < \varepsilon_1 = 1.$$
Choose $m = N_1$. For $n \geq N$, we have
$$|a_{N_1} - a_n| < 1.$$
Note that
$$|a_n| - |a_{N_1}| \leq |a_n - a_{N_1}| < 1.$$
Thus, for all $n \geq N_1$,
$$|a_n| \leq 1 + |a_{N_1}|.$$
Choose $M = \max\{|a_1|, |a_2|, \ldots, |a_N|\}$. Thus, we have that (a_n) is bounded.
2. To show that (a_n) has a convergent subsequence, by the Bolzano-Weierstrass Theorem, there is a subsequence $\left(a_{n_k}\right)$ of (a_n) that converges.

   Let $\lim_k a_{n_k} = L_1$.
3. Claim that $\lim_n a_n = L_1$. Let $\varepsilon > 0$. There is $N_2 \in \mathbb{N}$ such that for all $k \in \mathbb{N}$, if $a_{n_k} \geq N_2$, then $|a_{n_k} - L_1 < \varepsilon/2$. Since (a_n) is a Cauchy sequence, there is $N_3 \in \mathbb{N}$ such that for all $m, n \in \mathbb{N}$, if $m, n \geq N_3$, then $|a_m - a_n| < \varepsilon/2$. Choose $N = \max\{N_2, N_3\}$. If $n \geq N$ and $n_k \geq N$, then
$$\begin{aligned}
|a_n - L| &= |a_n - a_{n_k}| \\
&\leq \underbrace{|a_n - a_{n_1}|}_{\text{Cauchy}} + |a_{n_k} - L_1| \\
&< \varepsilon/2 + \varepsilon/2 \\
&= \varepsilon.
\end{aligned}$$
Thus, $\lim_n a_n = L_1$.

□

Fun Fact: Augustin Louis Cauchy

Cauchy is a major figure in the history of many branches of mathematics – number theory, the theory of finite groups, and more – but he is most widely known for his contributes to analysis, especially complex analysis.

Figure 2.1: Augustin Louis Cauchy [*Credit: Wikipedia*]

He is credited with inventing the ϵ-based definition of limits we use today. Although there were a few gaps in his work, it does not diminish his brilliance in any way. Cauchy pretty much laid the groundwork for modern standards of rigor in analysis.

Chapter 3

Basic Topology of $\mathbb{R}$

Topology is considered to be "rubber geometry", as it gives us a flexible framework to explore of a lot of ideas. For the purposes of this class, we will take a look at *point set topology*. To begin, consider the intervals $(0,1)$ and $[0,1]$, which we define to be open and closed, respectively. However, what does it really mean for an interval or set to be open or closed? This is what we will attempt to answer in this chapter.

3.1 Cantor Sets

To begin extending our intuition about the nature of subsets in the real line, we will take a look at the construction of the *Cantor Set*. Let C_0 be the closed interval $[0,1]$, and define C_1 to be the set that results when the open middle third is removed; that is

$$C_1 = C_0 \setminus \left(\frac{1}{3}, \frac{2}{3}\right) = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$

Construct C_2 similarly by removing the middle third of each component of C_1

$$C_2 = \left(\left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right]\right) \cup \left(\left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]\right).$$

We can continue this process inductively

Name	Set	Length	Intervals
C_0	$[0,1]$	1	1
C_1	$[0,\frac{1}{3}]\cup[\frac{2}{3},1]$	$\frac{2}{3}$	2
C_2	$[0,\frac{1}{9}]\cup[\frac{2}{9},\frac{1}{3}]\cup[\frac{2}{3},\frac{7}{9}]\cup[\frac{8}{9},1]$	$\frac{4}{9}$	4
C_3	$\cdots$	$\left(\frac{2}{3}\right)^3$	8
$\vdots$	$\vdots$	$\vdots$	$\vdots$
C_n	$\cdots$	$\left(\frac{2}{3}\right)^n$	2^n

Table 3.1: Cantor sets

Eventually, we get the set C_n consisting of 2^n closed intervals each having length $1/3^n$. We

define the *Cantor Set* to be

$$C = \bigcap_{n=0}^{\infty} C_n.$$

There might be some doubt as to whether or not anything even remains after taking the infinite intersection. We can see that

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \dots,$$

and we also note that C_n is a union of *closed* intervals. Thus, by the Nested Interval Property

$$\bigcap_{n\in\mathbb{N}} C_n \neq \emptyset.$$

This can also be determined from the fact that our endpoints $0, 1 \in C_n$ for each $n \in \mathbb{N}$. Is $C := \bigcap_{n\in\mathbb{N}} C_n$ finite? No. Is it countable? As it turns out, no as well. The set C actually has cardinality equal to that of $\mathbb{R}$. However, from the point of length, C measures the same size as a single point. It was probably the first example of fractal sets, as C has a non-integer/fractional dimension.

3.2 Open and Closed Sets

Given $a \in \mathbb{R}$ and $\epsilon > 0$, recall that the ϵ-neighborhood of a is the set

$$V_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\}.$$

In other words, $V_\epsilon(a)$ is the open interval $(a - \epsilon, a + \epsilon)$, centered at a with radius ϵ.

Definition . *A set $O \subseteq \mathbb{R}$ is an open set if for any $a \in O$, there is $\varepsilon > 0$ such that $V_\varepsilon \subseteq O$.*

Let's determine whether some sets are open or not.

Example 7. *Prove the set $A = (0, 1)$ is an open set.*

Proof. Let $a \in A$. We need to choose some ε such that $V_\varepsilon(a) \subseteq A$. Choose $\varepsilon = \min\{a, 1 - a\}$. Thus, we have the ε-neighborhood

$$V_\varepsilon(a) = (a - \varepsilon, a + \varepsilon) \subseteq (0, 1).$$

This is because we have chosen ε in a way that ensures $a - \varepsilon \geq 0$ and $a + \varepsilon \leq a + (a - a) \leq 1$. Thus, we conclude that $(0, 1)$ is an open set. □

Example 8. *Prove the set $\emptyset$ is an open set.*

Proof. Since $a \in \emptyset$ is false, $\emptyset$ satisfies the definition of an open set, i.e. if $a \in \emptyset$, then there is $\varepsilon > 0$ such that $V_\varepsilon(a) \subseteq \emptyset$. Thus, $\emptyset$ is an open set. □

Example 9. *Is $[0, 1)$ an open set?*

Let $a = 0$. For any $\varepsilon > 0$, we have

$$0 - \frac{\varepsilon}{2} > 0 - \varepsilon.$$

Thus,

$$0 - \frac{\varepsilon}{2} \in (0-\varepsilon, 0+\varepsilon) = V_\varepsilon(0).$$

However, $-\frac{\varepsilon}{2} \notin [0,1)$ for any $\varepsilon > 0$. Thus, $[0,1)$ is not an open set.

Example 10. *Is $(0,1) \cup (2,3)$ an open set?*

We can write a proof sketch. Let $a \in A = (0,1) \cup (2,3)$. Thus, $a \in (0,1)$ or $a \in (2,3)$. If $a \in (0,1)$ there is a neighborhood $V_\varepsilon(a) \subseteq (0,1)$ for some $\varepsilon > 0$. Similarly, we have the same reasoning for $(2,3)$. Therefore, $(0,1) \cup (2,3)$ is an open set.

More generally, we can show that (a,b) is an open set by choosing a point $x \in (a,b)$ and $\varepsilon = \min\{x-a, b-x\}$. We can extend this to arbitrary unions as well.

Theorem 3.2.3.

(i) The union of an arbitrary collection of open sets is open.

(ii) The intersection of a finite collection of open sets is open.

Proof.

1. Let $a \in \bigcup_{\lambda \in \Lambda} O_\lambda$. By definition of union, there is $\lambda_0 \in \Lambda$ such that $a \in O_{\lambda_0}$. Since O_{λ_0} is open, there is $\varepsilon > 0$ such that $V_\varepsilon(a) \subseteq O_{\lambda_0} \subseteq \bigcup_{\lambda \in \Lambda}$. Thus, $\bigcup_{\lambda \in \Lambda} O_\lambda$ is open. Thus, we have shown that an arbitrary union of open sets is open.
2. $\bigcap_{i=1}^n O_i = \emptyset$, then we are done. Otherwise, assume that $\bigcap_{i=1}^n O_i \neq \emptyset$. Let $a \in \bigcap_{i=1}^n O_i$. Thus, for $i = 1, 2, \ldots, n$, we have $a \in O_i$. Since O_i is open for each $i \in \{1, 2, \ldots, n\}$, there is $\varepsilon_i > 0$ such that $V_{\varepsilon_i}(a) \subseteq O_i$ for each $i \in \{1, 2, \ldots, n\}$. Choose $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n\}$. Since $\varepsilon_i > 0$ for each i, we have $\varepsilon > 0$. Thus, $V_\varepsilon(a) \subseteq O_i$ for each $i \in \{1, 2, \ldots, n\}$. We can conclude that $\bigcap_{i=1}^n O_i$ is open.

□

Example 11. For each $i \in \mathbb{N}$, let $O_i = \left(-\frac{1}{i}, \frac{1}{i}\right)$. We have that

$$\bigcap_{i \in \mathbb{N}} O_i = \{0\},$$

which is not open for sure, as we cannot construct an ε-neighborhood for a single point. However, something like $O_i = (0, \frac{1}{i})$ would have

$$\bigcap_{i \in \mathbb{N}} O_i = \emptyset,$$

which is open.

Closed Sets

Definition 3.4.2. *A point x is a limit point of a set A if every ϵ-neighborhood $V_\epsilon(x)$ of x intersects the set A at some point other than x.*

Limit points are also called *cluster points* or *accumulation points*. To put their definition simply, if x is a limit point of a set A, x is the limit of a sequence in A.

Theorem 3.2.5. *Let $A \subseteq \mathbb{R}$. A point $x \in \mathbb{R}$ is a limit point of A if and only if $x = \lim x_n$ for some sequence (x_n) contained in A satisfying $x_n \neq x$ for all $n \in \mathbb{N}$.*

Why do we have the restriction that $x_n \neq x$? Given a point $a \in A$, it is always the case that a is the limit of a sequence in A if we are allowed to consider the constant sequence $(a, a, a, \dots)$. There will be occasions where we want to avoid this uninteresting situation, so it is import to be able to distinguish limit points from *isolated points*.

Definition 3.2.6. *A point $a \in A$ is an isolated point of A if it is not a limit point of A.*

Let's find the limit points for a set.

Example 12. *What are the limit points of $A = (0, 1)$?*

In the case where $x = 0$, we let $a_n = \frac{1}{n+1}$. For each $n \in \mathbb{N}$, $a_n \in A$ and $\lim_n a_n = 0$. Thus, 0 is a limit point of A. In the case where $x = 1$, let $a_n = 1 - \frac{1}{n+1} \in A$ for each $n \in \mathbb{N}$. Thus, 1 is a limit point of A. In the case where $0 < x < 1$, for some $x \in A$, let $x_n = x - \frac{x}{n+1}$ for each $n \in \mathbb{N}$. Note that

$$\begin{aligned} x_n &= x - \frac{x}{n+1} \\ &= \frac{(n+1)x - x}{n+1} \\ &= \frac{nx}{n+1} \\ &= \frac{n}{n+1} \cdot x \\ &< \frac{n}{n+1} \cdot 1 \\ &< 1. \end{aligned}$$

Thus $\lim_n x_n = \lim_n \left(x - \frac{x}{n+1}\right) = \lim_n x - \lim_n \frac{x}{n+1} = x$ and x is a limit point of A. In the case that $x < 0$, $|0 - x| = |x| > 0$, we can choose $\varepsilon = \frac{|x|}{2} > 0$. Any sequence (x_n) in A has the property that

$$\begin{aligned} |x_n - x| &> |0 - x| \\ &= |x| \\ &> \frac{|x|}{2} \\ &= \varepsilon. \end{aligned}$$

Thus x is not a limit point of A. Similarly, $x > 1$ is not a limit point of A. We can conclude that the set of all limit points of A is $[0, 1]$.

Note that the set of limit points of $(a,b),[a,b),(a,b],[a,b]$ is $[a,b]$. We can now define what it means for a set to be closed.

Definition 3.2.7. *A set $F \subseteq \mathbb{R}$ is closed if it contains all its limit points.*

Closure

Definition 3.2.11. *Given a set $A \subseteq \mathbb{R}$, let L be the set of all limit points of A. The closure of A is defined to be $\overline{A} = A \cup L$.*

Complements

The mathematical notations of open and closed are not antonyms in the way that they are in standard English. Many sets such as the half-open interval $(c,d] = \{x \in \mathbb{R} : c < x \leq d\}$ are neither open nor closed. The sets $\mathbb{R}$ and $\emptyset$ are both simultaneously open and closed*. There is, however, an open relationship between open and closed sets. Recall that the *complement* of a set $A \subseteq \mathbb{R}$ is defined to be the set

$$A^c = \{x \in \mathbb{R} : x \notin A\}.$$

The relationship is the following

Theorem 3.2.13. *A set O is open if and only if O^c is closed. Likewise, a set F is closed if and only if F^c is open.*

Now that we know all definitions, let's show some examples of closed sets.'

Example 13. *Prove that $\{0\}$ is a closed set.*

To prove that $\{0\}$ is a closed set, we consider its complement

$$\{0\}^C = (-\infty, 0) \cup (0, \infty).$$

Since $\{0\}^C$ is a union of open sets, it is open, thus $\{0\}$ is closed.

Example 14. *Prove that $[0,1]$ is closed.*

Consider the complement of $[0,1]$

$$[0,1]^C = (-\infty, 0) \cup (1, \infty).$$

For similar reasons above, the complement of $[0,1]$ is open, thus $[0,1]$ is closed.

Example 15. *Is $[0,1)$ closed?*

Taking the complement of $[0,1)$, we have

$$[0,1]^C = \underbrace{(-\infty, 0)}_{\text{open}} \cup \underbrace{[1, \infty)}_{\text{not open}}.$$

*Thankfully, these are the only sets with this property

Example 16. *Is $\emptyset$ closed?*

The complement is $\emptyset^C = \mathbb{R}$, which is open, thus $\emptyset$ is closed.

Example 17. *What about $\mathbb{R}$?*

$\mathbb{R}^C = \emptyset$ which is open, thus $\mathbb{R}$ is closed.

So far, we have generalized the notion of open intervals to open sets and closed intervals to closed sets. What about compact sets?

3.3 Compact Sets

One important tool when utilizing the power of the mathematical infinite are *compact sets*, which often bring a more finite quality to many proofs.

Definition 3.3.1 (Compactness). *A set $C \subseteq \mathbb{R}$ is called a compact set if every sequence (x_n) in C has a convergent subsequence (x_{n_k}) such that $\lim_k x_{n_k} = x$ and $x \in C$.*

For example, let us show that $\{0\}$ is a compact set. Let (x_n) be a sequence in $\{0\}$. We have no choice but to choose $(x_n) = (0, 0, 0, \dots)$. Thus, $x_n = 0$ for all $n \in \mathbb{N}$. We have that $\lim_n x_n = 0$. Let (x_{n_k}) be a convergent subsequence of (x_n) such that $\lim_k x_{n_k} = x$. Since (x_{n_k}) is a subsequence of the convergent sequence (x_n) and $\lim_n x_n = 0$, it must be the case that $\lim_k x_{n_k} = 0$ as well. Thus $x = 0 \in \{0\}$.

Theorem 1. *Let F be a finite set. Then F is closed, i.e., $\overline{F} = F$.*

Proof. Let $\overline{F} = F \cup L_F$, where L_F is the set of limit points of F. If $F = \emptyset$, we are done. Assume F is non-empty. We will show that $L_F = \emptyset$. Assume to the contrary that there is $x \in L_F$. There is a sequence (x_n) in F such that $\lim_n x_n = x$ and $x_n \neq x$ for all $n \in \mathbb{N}$. Thus, $x_n - x > 0$ for all $n \in \mathbb{N}$. We have that $\{x_n \in F \mid n \in \mathbb{N}\} \subseteq F$. Since F is finite, say $F = \{a_1, a_2, \dots, a_k\}$, we have for each $n \in \mathbb{N}$, $x_n = a_{i_n} \in F$ for some $1 \leq i_n \leq k$. Choose $\varepsilon_0 = \min\{|x - x_n| \mid n \in \mathbb{N}\}$. We have $\varepsilon_0 > 0$ because $\{|x - x_n| \mid n \in \mathbb{N}\} \subseteq \{|x - a_i| \mid i = 1, 2, \dots, k\}$, which is finite. Thus, $|x - x_n| \geq \varepsilon_0 > 0$ for each $n \in \mathbb{N}$. Thus, (x_n) does not converge to x, a contradiction. Thus, $L_F = \emptyset$, i.e. the set of limit points is empty. $\square$

Note that any nonempty finite set is bounded.

Chapter 4

Functional Limits and Continuity

4.2 Functional Limits

The first type of functions we covered were sequences. We defined (a_n) to be a sequence on $\mathbb{R}$ as $a : \mathbb{N} \to \mathbb{R}$. Today we will be extending the domain from $\mathbb{N}$ to $\mathbb{R}$, i.e. $f : D \to \mathbb{R}$, where $D \subseteq \mathbb{R}$. Consider the function $f(x) = 2x + 1$, where $f : \mathbb{R} \to \mathbb{R}$. We know from Calculus 1 that

$$\lim_{x \to \frac{1}{2}} f(x) = 2\left(\frac{1}{2}\right) + 1 = 2.$$

In Calculus 1, we assumed that every function is nice (i.e. continuous) so substitution would work. Consider another function $g : \mathbb{R} \to \mathbb{R}$, where

$$g(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{Q}^C \end{cases}.$$

If we wanted to compute $\lim_{x \to 0} g(x)$, what would it be? Does it exist? We'll try to answer this question. Let's begin by rigorously defining what the limit of a function is.

Definition 4.2.1 (Functional Limits). *Let $f : A \to \mathbb{R}$, and let c be a limit point of the domain A. We say that $\lim_{x \to c} f(x) = L$ provided that, for all $\epsilon > 0$, there exists a $\delta > 0$ such that whenever $0 < |x - c| < \delta$ (and $x \in A$) it follows that $|f(x) - L| < \epsilon$.*

Visually, this definition implies the situation in Figure 4.1.

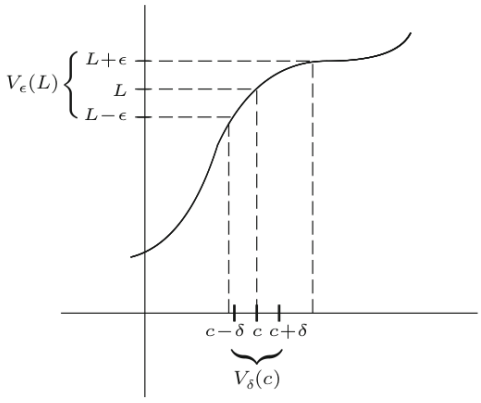


Figure 4.1: Functional Limit Definition

Let's show how to use this definition to prove that a function converges to a limit.

Example 18. Let $f(x) = 2x+1$. Does a limit of $f(x)$ at $x = 1$ exist? We claim that $\lim_{x\to 1} f(x) = 3$. Let $\epsilon > 0$. We want to find $\delta > 0$ such that if $0 < |x-1| < \delta$, then $|f(x) - 3| < \epsilon$. We have that

$$\begin{aligned} |f(x) - 3| &= |(2x+1) - 3| \\ &= |2x - 2| \\ &= 2|x-1|. \end{aligned}$$

Choose $\delta = \frac{\epsilon}{2} > 0$. Let $x \in \mathbb{R}$ such that $0 < |x-1| < \delta = \frac{\epsilon}{2}$. Then

$$\begin{aligned} |f(x) - 3| &= |(2x+1) - 3| \\ &= |2x - 2| \\ &= 2|x-1| \\ &< 2\delta \\ &= 2\left(\frac{\epsilon}{2}\right) \\ &= \epsilon. \end{aligned}$$

Thus, for any $\epsilon > 0$, there is $\delta = \frac{\epsilon}{2} > 0$, such that $0 < |x-1| < \delta$ implying $|f(x) - 3| < \epsilon$. We can conclude that $\lim_{x\to 1} f(x) = 3$.

With this new concept of functional limits in mind, we can return to our function

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{Q}^C \end{cases}.$$

Does the limit of $f(x)$ exist at $x = 0$? We will show that it does not. Assume to the contrary that $\lim_{x\to 0} f(x) = L$ for some $L \in \mathbb{R}$. Thus, for every $\epsilon > 0$, there is $\delta > 0$ such that $0 < |x - 0| < \delta$, implying $|f(x) - L| < \epsilon$. Let $\epsilon = \frac{1}{2}$. Thus, there is $\delta_1 > 0$ such that $0 < |x - 0| < \delta_1$ implies $|f(x) - L| < \epsilon = \frac{1}{2}$. Let $x \in (-\delta_1, \delta_1) \setminus \{0\}$.

- If $x \in \mathbb{Q}$, then $f(x) = 1$, by its definition. Thus $|f(x) - L| = |1 - L| < \epsilon = \frac{1}{2}$, i.e. $|1 - L| < \frac{1}{2}$.

- If $x \notin \mathbb{Q}$, then $f(x) = 0$. Thus, $|0 - L| < \frac{1}{2}$.

We have from case 1 that

$$|1 - L| < \frac{1}{2} \implies 1 + \frac{1}{2} > L > 1 - \frac{1}{2},$$

and from case 2 we have that

$$|L| < \frac{1}{2} \implies -\frac{1}{2} < L < \frac{1}{2}.$$

From cases 1 and 2, we have that $L > \frac{1}{2}$ and $L < \frac{1}{2}$. This is, of course, a contradiction. Thus, $f(x)$ does not have a limit at $x = 0$. In fact, $f(x)$ does not have a limit at $x = a$, for each $a \in \mathbb{R}$.

Sequential Criterion for Functional Limits

In Chapter 2, we derived a list of properties held by sequential limits (in particular, the Algebraic Limit Theorem and the Order Limit Theorem). As it turns out, there are analogous statements for functional limits. Although it is not difficult to prove each individual property, all of them will following natually from their sequential counterparts once we derive the sequential criterion for functional limits.

Theorem 4.2.3 (Sequential Criterion for Functional Limits). *Given a function $f : A \to \mathbb{R}$ and a limit point c of A, the following two states are equivalent:*

(i) $\lim_{x \to a} f(x) = L$.

(ii) For all sequences $(x_n) \subseteq A$ satisfying $x_n \neq c$ and $(x_n) \to c$, it follows that $f(x_n) \to L$.

Proof.

(1) $\implies$ (2).

Assume that $\lim_{x \to c} f(x) = L$. Let (a_n) be a sequence in A such that $\lim a_n = c$. We will show that $(f(a_n))$ converges to L. Let $\epsilon > 0$. There is $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$. Since $\lim a_n = c$, there is $N \in \mathbb{N}$ such that for every $n \in \mathbb{N}$, if $n \geq N$, then $0 < |a_n - a| < \delta$. Therefore, by the same reasoning, $|f(a_n) - L| < \epsilon$.

(2) $\implies$ (1)

Assume the "sequential criterion". Assume to the contrary that $\lim_{x \to a} f(x) \neq L$. Thus, there is $\epsilon_0 > 0$ such that for any $\delta > 0$, there is $x \in \mathbb{R}$ such that $0 < |x - a| < \delta$ but $|f(x) - L| \geq \epsilon_0$. In terms of logic, we have

$$(\forall \epsilon > 0, \exists \delta > 0, 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$

We can turn this into an if P then Q statement

$$\forall \epsilon > 0, \exists \delta > 0 (\neg P \vee Q).$$

Negating this, we obtain

$$\exists \epsilon > 0, \forall \delta > 0 (P \wedge \neg Q).$$

Thus, we have that for each $\delta_n = \frac{1}{n} > 0$ where $n \in \mathbb{N}$, there is x_n such that $0 < |x_n - a| < \delta_n = \frac{1}{n}$ and/but $|f(x_n) - L| \geq \epsilon_0$. Thus, $(f(x_n))$ does not converge to L, a contradiction. □

Corollary 4.2.4 (Algebraic Limit Theorem for Functional Limits). *Let f and g be functions defined on a domain $A \subseteq \mathbb{R}$, and assume $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$ for some*

limit point c of A. Then,

(i) $\lim_{x\to c} kf(x) = kL$ *for all* $k \in \mathbb{R}$

(ii) $\lim_{x\to c}[f(x)+g(x)] = L+M$

(iii) $\lim_{x\to c}[f(x)g(x)] = LM$

(iv) $\lim_{x\to c} f(x)/g(x) = L/M$, *provided* $M \neq 0$.

Remark 1. *Note that the limit of a function is unique, i.e., if* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} f(x) = M$, *then* $L = M$.

Proof. Let $f : A \to \mathbb{R}$ and let c be a limit point of A. We have that for a point $x \in A$, $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} f(x) = M$ if for every $\epsilon > 0$, there exist $\delta_1, \delta_2 > 0$ such that $0 < |x-c| < \delta_1$ implies that $|f(x) - L| < \epsilon$ and $0 < |x-c| < \delta_2$ implies that $|f(x) - M| < \epsilon$. Choose $\delta = \min\{\delta_1, \delta_2\} > 0$. There exists $x \in A$ such that whenever $0 < |x-a| < \delta$

$$|L-M| = |L - f(x) + f(x) - M| = |L - f(x)| + |f(x) - M| < \epsilon.$$

As ϵ was chosen arbitrarily, we must have $L = M$. □

4.3 Continuous Functions

Next up we have a significant milestone in our journey toward a rigorous theory of real-valued functions – a proper definition of continuity.

Definition 4.3.1 (Continuity). *A function $f : A \to \mathbb{R}$ is continuous at a point $c \in A$ if for all $\epsilon > 0$, there exists a $\delta > 0$ such that whenever $|x-c| < \delta$[a] (and $x \in A$), it follows that $|f(x) - f(c)| < \epsilon$. If f is continuous at every point in the domain A, then we say that f is continuous on A.*

[a] Note that we don't include $0 < |x-a|$ because we allow $x = a$.

The definition of continuity is quite similar to the definition for functional limits, however, there are a few subtle differences. The most important difference is that we require the point c to be in the domain of f. The next example illustrates how to prove a function is continuous at a point.

Example 19. Prove that $f : (0,2) \to \mathbb{R}$ defined by $f(x) = x^2 + 1$ is continuous at $x = 1$.

Proof. Observe first that

$$f(1) = (1)^2 + 1 = 2.$$

Let $\epsilon > 0$. We want to find $\delta > 0$ such that $|x-1| < \delta$ implies

$$|f(x) - f(1)| = |(x^2+1) - 2| = |x^2 - 1| = |x+1|\underbrace{|x-1|}_{<\delta}.$$

We want to bound $|x+1|$. Since $x \in (1-\delta, 1+\delta)$, then $x+1 \in (-\delta, 2+\delta)$ and we have $|x+1| < 2+\delta$. Observe that

$$\begin{aligned}\delta(\delta+2) &< \delta(1+2)\\ &= 3\delta\\ &= 3\left(\frac{\epsilon}{3}\right)\\ &< \epsilon.\end{aligned}$$

Thus, we choose $\delta = \min\{1, \frac{\epsilon}{3}\}$. Now we have

$$\begin{aligned}|f(x)-2| = |x-1||x+1| &< \delta(2+\delta)\\ &< 3\delta\\ &\leq 3\left(\frac{\epsilon}{3}\right)\\ &< \epsilon.\end{aligned}$$

□

Naturally, we also have an Algebraic Continuity Theorem.

Theorem 4.3.4 (Algebraic Continuity Theorem). *Let $f : A \to \mathbb{R}$ and $g : A \to \mathbb{R}$ be continuous functions at a point $c \in A$ and $k \in \mathbb{R}$. Then*

(i) $kf(x)$ is continuous at c for all $k \in \mathbb{R}$.

(ii) $f(x) \pm g(x)$ is continuous at c.

(iii) $f(x)g(x)$ is continuous at c.

(iv) $f(x)/g(x)$ is continuous at c, provided the quotient is defined.

Proof.

(i)

Can be proved using $\epsilon - \delta$ or sequential definition. We will use the latter. Let (x_n) be a sequence in A such that $\lim_n x_n = a$. Since f is continuous at $x = c$, then $\lim_n f(x_n) = f(c)$. Thus, $\lim_n kf(x_n) = kf(c)$ for $k \in \mathbb{R}$. Therefore, $kf(x)$ is continuous at $x = c$.

(ii), (iii), (iv)

Left as an exercise to the reader. □

Example 20. Let us revisit our earlier example $f : (0,2) \to \mathbb{R}$ defined by $f(x) = x^2 + 1$. Prove that f is continuous at $x = 1$. We can show that

$$f(x) = (x)(x) + 1 = f_1(x)f_2(x) + g(x),$$

and prove that each is continuous. For $f(x) = x$, let $\epsilon > 0$. We choose $\delta = \epsilon$. If $|x-1| < \delta = \epsilon$, then $|f(x) - f(1)| = |x-1| < \delta = \epsilon$. Thus, $f(x) = x$ is continuous at $x = 1$. For $g(x) = 1$, let $\epsilon > 0$, choose δ to be anything positive number. If $|x-1| < \delta$, then $|f(x) - f(1)| = |1-1| = 0 < \epsilon$. Thus, $g(x) = 1$ is continuous at $x = 1$. By Theorem 4.3.4, we have that $f(x)f(x) = x^2$ is continuous at $x = 1$, as is $x^2 + 1$.

4.4 Continuous Functions on Compact Sets

One interesting question to ask about a function is which properties are preserved when a particular set E is mapped to $f(E)$ via a continuous function. For example, if E is open and f is continuous, is $f(E)$ necessarily open? The answer to this question is no – if $f(x) = x^2$ and E is the open interval $(-1,1)$, then $f(E)$ is the interval $[0,1)$. It turns out to be false for closed sets as well. Consider the function

$$g(x) = \frac{1}{1+x^2},$$

and the closed set $E = [0,\infty)$. Because $g(E) = (0,1]$ is not closed, we must conclude that continuous functions do not, in general, map closed sets to closed sets. Note however, that the previous counterexamples required the use of open or unbounded sets. This is not a coincidence. Sets that are closed and bounded – i.e., compact sets – always get mapped to closed and bounded subsets by continuous functions.

Theorem 4.4.1 (Preservation of Compact Sets). *Let $f : A \to \mathbb{R}$ be continuous on A. If $E \subseteq A$ is compact, then $f(E)$ is compact as well.*

Proof (Using Sequential Compactness). In the case that $E = \emptyset$, then $f(E) = f(\emptyset) = \emptyset$. Consider $E \neq \emptyset$. We want to show that $f(E)$ is compact. Let (y_n) be an arbitrary sequence in $f(E)$. For each $n \in \mathbb{N}$, choose $x_n \in E$ such that $f(x_n) = y_n$. Therefore, (x_n) is a sequence in E. Since E is compact, there is a convergent subsequence, say (x_{n_k}) in E such that $\lim_k x_{n_k} = x \in E$. Since f is continuous, then $\lim_k y_{n_k} = \lim_k f(x_{n_k}) = f(x) \in f(E)$. Hence $f(E)$ is compact. □

One extremely important corollary is obtained by combining this result with the observation that compact sets are bounded and contain their supremums and infimums.

Theorem 4.4.2 (Extreme Value Theorem). *If $f : K \to \mathbb{R}$ is continuous on a compact set $K \subseteq \mathbb{R}$, then f attains a maximum and minimum value. In other words, there exist $x_0, x_1 \in K$ such that $f(x_0) \leq f(x) \leq f(x_1)$ for all $x \in K$.*

Proof. Assume K is compact and $f : K \to \mathbb{R}$ is continuous. Thus, $f(K)$ is compact. By the Heine-Borel Theorem, $f(K)$ is closed and bounded. Since $f(K)$ is bounded, $\inf f(K)$ and $\sup f(K)$ exist. Since $f(K)$ is closed, $\inf f(K) \in f(K)$ and $\sup f(K) \in f(K)$. Thus, $\inf f(K)$ is a minimum and $\sup f(K)$ is a maximum. □

Uniform Continuity

Today, we will introduce to concept of functions that are everywhere continuous, i.e., uniform continuity.

Definition 4.4.4 (Uniform Continuity). *A function $f : A \to \mathbb{R}$ is uniformly continuous on A if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in A$, $|x-y| < \delta$ implies $|f(x) - f(y)| < \epsilon$.*

For example, we can show that the function $f(x) = x$ is continuous on $\mathbb{R}$.

Example 21. *Let $f(x) = x$ where $f : \mathbb{R} \to \mathbb{R}$. Is f uniformly continuous on $\mathbb{R}$?*

As f is already continuous on $\mathbb{R}$, it should be the case that it is uniformly continuous. We

will prove this rigorously. Let $x, c \in \mathbb{R}$ and $\epsilon > 0$. We want to find $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon$. Choose $\delta = \epsilon > 0$. If $|x - c| < \delta$, then $|f(x) - |f(c)| = |x - c| < \epsilon$.

Let's contrast this with what happens when we try to prove $g(x) = x^2$ is continuous on $\mathbb{R}$.

Example 22. *Let $f(x) = x^2$ where $f : \mathbb{R} \to \mathbb{R}$. Is $f(x)$ uniformly continuous?*

Let $\epsilon > 0$. In terms of logic, we have

$$\forall \epsilon > 0, \forall x, c \in \mathbb{R}, \exists \delta > 0, |x - c| < \delta \implies |f(x) - f(c)| < \epsilon.$$

If we negate this, we obtain

$$\exists \epsilon > 0, \exists x, c \in \mathbb{R}, \forall \delta > 0, |x - c| < \delta \wedge |f(x) - f(c)| \geq \epsilon.$$

Note that this is because the $|x - c| < \delta \implies |f(x) - f(c) < \epsilon$ is in the form $P \implies Q \equiv \neg(P \wedge \neg Q)$. Choose $x, c \in \mathbb{R}$ such that $|x - c| = \frac{\delta}{2}$ and $|x + c| > \frac{2\epsilon}{\delta}$. This is because for the negation, we have

$$\begin{aligned} |f(x) - f(c)| &\geq \epsilon \\ |x^2 - c^2| &\geq \epsilon \\ |x - c||x + c| &\geq \epsilon. \end{aligned}$$

Thus

$$\begin{aligned} |f(x) - f(c)| &= |x^2 - c^2| \\ &= |x - c||x + c| \\ &> \frac{\delta}{2} \cdot \frac{2\epsilon}{\delta} \\ &= \epsilon. \end{aligned}$$

From this, we have that f is not uniformly continuous on $\mathbb{R}$.

Once again, if we assume that the domain is compact, we have a nice result in terms of continuity.

Theorem 4.4.7 (Uniform Continuity on Compact Sets). *A function that is continuous on a compact set K is uniformly continuous on K.*

Proof. Assume that $f : K \to \mathbb{R}$ is continuous and K is compact. Let $\epsilon > 0$. Since f is continuous for any $x \in A$, there is $\delta_x > 0$ such that $|x - c| < \delta_x$ implies $|f(x) - f(c)| < \frac{\epsilon}{2}$. For any $x \in A$, define $O_\epsilon(x) = \{c \in A \mid |x - c| < \delta_x\}$. Note that $O_\epsilon(x)$ is open and $|f(x) - f(c)| < \epsilon$ for any $c \in O_\epsilon(x)$. Let $U = \{O_\epsilon(x) \mid x \in A\}$ be an open cover of A. Since A is compact, U has a finite subcover, say $U_n = \{O_\epsilon(x_i) \mid i = 1, 2, 3, \ldots. n\}$. Let $\delta_0 = \min\{\delta_{x_1}, \delta_{x_2}, \ldots, \delta_{x_n}\} > 0$. Now, let $\delta' = \min\{(x_i + \delta_{x_i}) - (x_j - \delta_{x_j}) > 0 \mid i \neq j, i, j \in \{1, 2, \ldots, n\}$. Let $\delta = \min\{\delta_0, \delta'\} > 0$. For any $x, c \in A$

such that $|x-c|<\delta$, we have $x,c \in O_\epsilon(x_{i_0})$ for some $i_0 = 1,2,\ldots,n$. Therefore,

$$\begin{aligned}|f(x)-f(c)| &= |f(x)-f_{x_{i_0}}+f(x_{i_o}-f(c)| \\ &\leq |f(x)-f(x_{i_0})|+|f(x_{i_o}-f(c)| \\ &< \frac{\epsilon}{2}+\frac{\epsilon}{2} \\ &= \epsilon.\end{aligned}$$

Thus, f is uniformly continuous on A. □

4.5 Intermediate Value Theorem

The Intermediate Value Theorem is the name given to the very intuitive observation that a continuous function f on a closed interval $[a,b]$ attains every value that falls between the range values $f(a)$ and $f(b)$.

Theorem 4.5.1 (Intermediate Value Theorem). *Let $f:[a,b]\to\mathbb{R}$ be continuous. If L is a real number satisfying $f(a)<L<f(b)$ or $f(a)>L>f(b)$, then there exists a point $k\in(a,b)$ where $f(k)=L$.*

Proof. Assume that $f:[a,b]\to\mathbb{R}$ is continuous and L is between $f(a)$ and $f(b)$. Without loss of generality, assume $f(a)\leq L\leq f(b)$. Since $[a,b]$ is connected, $f([a,b])$ is connected. Does $f([a,b])=[c,d]$ for some $c,d\in\mathbb{R}$? Yes, by the Heine-Borel theorem (since $[a,b]$ is closed and bounded). We have that $c\leq f(a)\leq L\leq f(b)\leq d$. Since $L\in[c,d]=f([a,b])$, there is $k\in[a,b]$ such that $f(k)=L$. □

The most potentially useful way to understand the Intermediate Value Theorem is as a special case of the fact that continuous functions map connected sets to connected sets.

Theorem 4.5.2 (Preservation of Connected Sets). *Let $f:G\to\mathbb{R}$ be continuous. If $E\subseteq G$ is connected, then $f(E)$ is continuous as well.*

Proof. Assume that f is continuous and E is connected. Let $f(E)=A\cup B$ where A and B are disjoint and nonempty, i.e., $A\cap B=\emptyset$ and $A\neq\emptyset\neq B$. We will next define

$$C=\{x\in E: f(x)\in A\},\quad D=\{x\in E: f(x)\in B\}.$$

The sets C and D are called the preimages of A and B, respectively. Using the properties of A and B, we see that C and D are nonempty and disjoint, and thus satisfy $E=C\cup D$. Assuming E is connected, there exists a sequence (x_n) in one of C or D with $x=\lim x_n$ contained in the other. Since f is continuous, $\lim_n f(x_n)=f(x)$. Thus, there exists a sequence $(y_n)=(f(x_n))$ in either A or B and $\lim_n y_n=f(x)$ is in the other. Thus, $f(E)$ is connected. □

Definition (One-Sided Limit). *We say that $\lim_{x\to c^+} f(x)=L$ if for all $\epsilon>0$, there exists $\delta>0$ such that whenever $0<x-a<\delta$, we have that $|f(x)-L|<\epsilon$.*

Similarly, we say that $\lim_{x\to c^-}=M$ if for all $\epsilon>0$, there exists $\delta>0$ such that whenever $0<a-x<\delta$, we have that $|f(x)-M|<\epsilon$.

Chapter 5

The Derivative

The Derivative

The theory of functional limits from Chapter 4 is all that is needed to supply a rigorous definition for the derivative.

Definition 5.2.1 (Differentiability). *Let $f : A \to \mathbb{R}$ be a function defined on an interval A. Given $c \in A$, the derivative of f and c is defined by*

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c},$$

provided that this limit exists. In this case, we say f is differentiable at c. If f' exists for all points $c \in A$, we say that f is differentiable on A.

We can show the usage of this definition with an example.

Example 23. Consider $f(x) = x^n$ for $n \in \mathbb{N}$, and let c be any arbitrary point in $\mathbb{R}$. Using the algebraic identity

$$x^n - c^n = (x - c)(x^{n-1} + cx^{n-2} + c^2x^{n-3} + \cdots + c^{n-1}),$$

we can calculate the familiar formula

$$\begin{aligned} f'(c) &= \lim_{x \to c} \frac{x^n - c^n}{x - c} \\ &= \lim_{x \to c} (x^{n-1} + cx^{n-2} + c^2x^{n-3} + \cdots + c^{n-1}) \\ &= c^{n-1} + c^{n-1} + \cdots + c^{n-1} \\ &= nc^{n-1}. \end{aligned}$$

Observe that the continuity of a function does not necessarily imply it is differentiable. On the other hand, if a function is differentiable at a point, it is true that it must be continuous at that point.

Theorem 5.2.3. *If $f : A \to \mathbb{R}$ is differentiable at a point $c \in A$, then f is continuous at c as well.*

Proof (Class). We want to show that f is continuous at $c \in A$. Assume $f' = L$. Let $\epsilon > 0$. There is $\delta > 0$ such that $0 < |x - c| < \delta$ implies

$$\left| \frac{f(x) - f(c)}{x - c} - L \right| < \epsilon.$$

We can rearrange as follows

$$|f(x) - f(c) - L(x - c)| < \epsilon |x - a|.$$

Choose δ_1 so that for $|x - a| < \delta_1$, we have $|x - a| < \frac{1}{2}$ and $|x - a| < \frac{\epsilon/2}{|L|}$, i.e., $\delta_1 = \min\left\{\frac{1}{2}, \frac{\epsilon/2}{|L|}, \right\}^*$. Thus, there is $\delta' = \min\{\delta, \delta_1\}$ such that $0 < |x - a| < \delta'$ implies

$$|f(x) - f(a) - L(x - a)| \geq |f(x) - f(a)| - |L(x - a)|.$$

Rearranging with our ϵ term, we have

$$\begin{aligned} |f(x) - f(a)| &< \epsilon |x - a| + |L||x - a| \\ &< \epsilon \left(\frac{1}{2}\right) + |L| \left(\frac{\epsilon/2}{L}\right) \\ &< \epsilon. \end{aligned}$$

Thus, f is continuous at $c \in A$. □

Proof (Book). We assume that

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c},$$

exists, and we want to prove that $\lim_{x \to c} f(x) = f(c)$. The Algebraic Limit Theorem for functional limits allow us to write

$$\lim_{x \to c} (f(x) - f(c)) = \lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) (x - c) = f'(c) \cdot 0 = 0.$$

It follows then that $\lim_{x \to c} f(x) = f(c)$. □

Combinations of Differentiable Functions

Theorem 5.2.4 (Algebraic Differentiability Theorem). *Let f and g be functions defined on an interval A, and assume both are differentiable at some point $c \in A$. Then,*

(a) $(f + g)'(c) = f'(c) + g'(c)$,

(b) $(kf)'(c) = kf'(c)$, *for all* $k \in \mathbb{R}$,

(c) $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$, *and*

(d) $(f/g)'(c) = \frac{g(c)f'(c) - f(c)g'(c)}{[g(c)]^2}$, *provided that* $g(c) \neq 0$.

*If $L = 0$, then $\frac{\epsilon/2}{|L|}$ does not exist.

5.1 The Mean Value Theorems

The Mean Value Theorem makes the assertion that a differentiable function f on an interval $[a,b]$ will, at some point, attain a slope equal to the slope of the line through the endpoints $(a, f(a))$ and $(b, f(b))$. In other terms,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for at least one point $c \in (a,b)$. Although this result is geometrically unsurprising, the Mean Value Theorem is the cornerstone of the proof for almost every major theorem pertaining to differentiation. One special case of the Mean Value Theorem first noted by the mathematician Michel Rolle is given below.

Definition 5.3.1 (Rolle's Theorem). *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). If $f(a) = f(b)$, then there exists a point $c \in (a,b)$ where $f'(c) = 0$.*

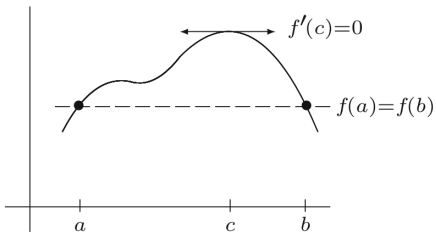


Figure 5.1: Rolle's Theorem

Proof. If f is a constant function, we are done. Assume that f is non-constant. Since f is differentiable on (a,b) it is continuous. By the Extreme Value Theorem, there is $c \in (a,b)$ such that $f(c)$ is a max or min. Note that such c can be chosen because f is non-constant[†]. Without loss of generality, assume that $f(c)$ is a maximum. We have that

$$\frac{f(x) - f(c)}{x - c} \geq 0,$$

for $x < c$[‡]. Similarly, we also have that for $y > c$

$$\frac{f(y) - f(c)}{y - c} \leq 0.$$

Thus, we have that

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \geq 0 \quad \text{and} \quad \lim_{y \to c^+} \frac{f(y) - f(c)}{y - c} \leq 0.$$

Since $f'(c)$ exists, it must be the case that

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} = \lim_{y \to c^+} \frac{f(y) - f(c)}{y - c}.$$

Thus, $f'(c) = 0$. □

[†]If both the maximum and minimum occur at the endpoints, then f is necessarily a constant function and $f'(x) = 0$ on all of (a,b).

[‡]This is because $f(x) - f(c) \leq 0$ since $f(c)$ is a maximum.

We can extend Rolle's Theorem to the Mean Value Theorem, which we will use to prove the Fundamental Theorem of Calculus.

Theorem 5.3.2 (Mean Value Theorem). *If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b), then there exists a point $c \in (a,b)$ where*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Observe that the Mean Value Theorem reduces to Rolle's Theorem in the case where $f(a) = f(b)$. The equation of the line through $(a, f(a))$ and $(b, f(b))$ (in point-slope form) is

$$y = \left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a).$$

We want to consider the difference between this line and the function $f(x)$. To this end, let

$$d(x) = f(x) - \left[\left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a)\right].$$

Observe that d is continuous on $[a,b]$, differentiable on (a,b), and satisfies $d(a) = 0 = d(b)$. By Rolle's Theorem, there exists a point $c \in (a,b)$ where $d'(c) = 0$. Because

$$d'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

we have that

$$0 = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

Rearranging, we see that

$$f'(c) = \frac{f(b) - f(a)}{b - a},$$

thus completing the proof. □

L'Hospital's Rules

The Algebraic Limit Theorem asserts that when taking a limit of a quotient of functions, we can write

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)},$$

provided that each individual limit exists and $\lim_{x \to c} g(x)$ is not zero. To handle the indeterminant cases when either the numerator and denominator both tend to zero or both tend simultaneously to infinity, we make use of L'Hospital's Rules. L'Hospital's rules are named for Marquis de L'Hospital, who learned the results from his tutor Johann Bernoulli, and published them in 1696.

Theorem 5.3.6 (L'Hospital's Rule: 0/0 case). *Let f and g be continuous on an interval containing a, and assume f and g are differentiable on this interval with the possible exception of*

the point a. If $f(a) = g(a) = 0$ and $g'(x) \neq 0$ for all $x \neq a$, then

$$\lim_{x\to a} \frac{f'(x)}{g'(x)} = L \implies \lim_{x\to a} \frac{f(x)}{g(x)} = L.$$

Proof. Assume that $f'(a)$ and $f'(b)$ exist. We have that

$$\begin{aligned}
\lim_{x\to a} \frac{f(x)}{g(x)} &= \lim_{x\to a} \frac{f(x)-f(a)}{g(x)-g(a)} \\
&= \lim_{x\to a} \left(\frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}} \right) \\
&= \left(\frac{\lim_{x\to a} \frac{f(x)-f(a)}{x-a}}{\lim_{x\to a} \frac{g(x)-g(a)}{x-a}} \right) \\
&= \frac{f'(a)}{g'(a)}.
\end{aligned}$$

□

Theorem 5.3.8 (L'Hospital's Rule: ∞/∞ case). *Assume f and g are differentiable on (a,b) and that $g'(x) \neq 0$ for all $x \in (a,b)$. If $\lim_{x\to a} g(x) = \infty$ or $(-\infty)$, then*

$$\lim_{x\to a} \frac{f'(x)}{g'(x)} = L \implies \lim_{x\to a} \frac{f(x)}{g(x)} = L.$$

Chapter 7

The Riemann Integral

7.2 The Definition of the Riemann Integral

The Riemann integral is the one usually discussed in introductory calculus. Starting with a function f on $[a,b]$ we partition the domain into small subintervals. We construct a row of rectangles to approximate the area between f and the x-axis. The total area of all the rectangles is given by the *Riemann sum*. On each subinterval $[x_{k-1}, x_k]$, we pick some point $c_k \in [x_{k-1}, x_k]$ and use the y-value $f(c_k)$ as an approximation for f on $[x_{k-1}, x_k]$. Thus, the area of each rectangle is $f(c_k)(x_k - x_{k-1})$ so the Riemann sum is

$$\sum_{k=1}^{n} f(c_k)(x_k - x_{k-1}).$$

To be able to prove the Fundamental Theorem of Calculus, we need to understand how Riemann integrals work.

Definition 7.2.1. *A partition P of $[a,b]$ is a finite set of points from $[a,b]$ that includes both a and b. The notational convention is to always list the points of a partition $P = \{x_0, x_1, x_2, \ldots, x_n\}$ in increasing order; thus*

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

For each interval $[x_{k-1}, x_k]$ of P, let

$$m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\}, \quad M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\}.$$

The lower sum of f with respect to P is given by

$$L(f,P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}).$$

Likewise, we define the upper sum of f with respect to P by

$$U(f,P) = \sum_{k=1}^{n} M_k(x_k - x_{k-1}).$$

Example 24. Let $f(x) = x$ on $[0,1]$ and $P = \{0, \frac{1}{2}, \frac{2}{3}\}$. We have that

$$\begin{aligned} m_1 &= \inf\left\{ f(x) \mid 0 \le x \le \frac{1}{2} \right\} \\ &= f(0) \\ &= 0. \end{aligned}$$

Similarly, we find that $m_2 = 1/2$ and $m_3 = 2/3$. Furthermore, we can calculate

$$\begin{aligned} M_1 &= \sup\{ f(x) \mid 0 \le x \le \frac{1}{2} \} \\ &= \frac{1}{2}. \end{aligned}$$

Performing similar calculations, we find that $M_2 = 2/3$ and $M_3 = 1$. Then

$$\begin{aligned} L(f,P) &= \sum_{k=1}^{n} m_k(a_k - a_{k-1}) \\ &= 0\left(\frac{1}{2} - 0\right) + \frac{1}{2}\left(\frac{2}{3} - \frac{1}{2}\right) + \frac{2}{3}\left(1 - \frac{2}{3}\right) \\ &= \frac{11}{36} \\ U(f,P) &= \sum_{k=1}^{3} M_k(a_k - a_{k-1}) \\ &= \frac{1}{2}\left(\frac{1}{2} - 0\right) + \frac{2}{3}\left(\frac{2}{3} - \frac{1}{2}\right) + 1 \cdot \left(1 - \frac{2}{3}\right) \\ &= \frac{25}{36}. \end{aligned}$$

We see that $L(f,P) \le U(f,P)$.

With this new concept of partitions in mind, we introduce another useful tool.

Definition 7.2.2 (Refinement). *A partition Q is a refinement of a partition P if Q contains all the points of P; that is, if $P \subseteq Q$.*

We can show this with an example.

Example 25. Let $[a,b] = [0,1]$ with partitions $P = \{0, 1/2, 2/3, 1\}$. An alternate partition $Q = \{0, 1/2, 2/3, 3/4, 5/6\}$ is a refinement of P, as $P \subseteq Q$.

Exploring refinements, we find some interesting results.

Lemma 7.2.3. *If Q is a refinement of P on $[a,b]$, then $L(f,P) \le L(f,Q) \le U(f,Q) \le U(f,P)$.*

Proof. Let $P = \{a_0 = a, a_1, \ldots, a_n = b\}$ and $Q = \{b_0 = a, b_1, \ldots, b_m = b$. Also, let $k = \{1, 2, \ldots, n\}$. We have that on the subinterval $[a_{k-1}, a_k]$, then there are $b_{k-1}^{(0)} = a_{k-1} < b_{k-1}^{(1)} < \cdots < b_{k-1} = a_k$, where

$b_{k-1}^{(j)} \in Q$ for $j = 0, \ldots, l$. For k, j

$$\inf\{f(x) \mid a_{k-1} \leq x \leq a_k\} \leq \inf\{f(x) \mid b_{k-1}^{(j)} \leq x \leq b_{k-1}^{(j+1)}\}$$

and

$$\sup\{f(x) \mid a_{k-1} \leq x \leq a_k\} \geq \sup\{f(x) \mid b_{k-1}^{(j)} \leq x \leq b_{k-1}^{(j+1)}\}.$$

Thus $L(f,P) \leq L(f,Q) \leq U(f,Q) \leq U(f,P)$. □

Lemma 7.2.4. *If P_1 and P_2 are any two partitions of $[a,b]$, then $L(f,P_1) \leq U(f,P_2)$.*

Proof. Let $Q = P_1 \cup P_2$ be the so called common refinement of P_1 and P_2. Because $P_1 \subseteq Q$ and $P_2 \subseteq Q$, it follows from Lemma 7.2.3 that

$$L(f,P_1) \leq L(f,Q) \leq U(f,Q) \leq U(f,P_2).$$

□

Definition 7.2.5. *Let $\mathcal{P}$ be the collection of all possible partitions of the interval $[a,b]$. The upper integral of f is defined to be*

$$U(f) = \inf\{U(f,P) : P \in \mathcal{P}\}.$$

In a similar fashion, we define the lower integral of f by

$$L(f) = \sup\{L(f,P) : P \in \mathcal{P}\}.$$

Note that we are able to find the infimum and supremum as the set of partitions is not empty and the function f is bounded.

Lemma 7.2.6. *For any bounded function f on $[a,b]$, it is always the case that $U(f) \geq L(f)$.*

Proof. For any partition P on $[a,b]$, we have that $L(f,P) \leq U(f,P)$. For any partition Q on $[a,b]$, $L(f,P) \leq U(f,Q)$ and $L(f,Q) \leq U(f,P)$. We have that

$$L(f) = \sup\{L(f,P) \mid P \text{ is a partition}\} \leq U(f,Q).$$

Similarly

$$L(f) \leq \inf\{U(f,Q) \mid Q \text{ is a partition}\} = U(f).$$

Thus, $L(f) \leq U(f)$. □

Definition 7.2.7 (Riemann Integrability). *A bounded function f defined on the interval $[a,b]$ is Riemann-integrable if $U(f) = L(f)$. In this case, we define $\int_a^b f$ or $\int_a^b f(x)\,dx$ to be this common value; namely,*

$$\int_a^b f = U(f) = L(f).$$

Example 26. The Dirchlet function $f : [0,1] \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases}$$

Let $P = \{a_0 = 0, a_1, \dots, a_n = 1\}$ be a partition on $[0,1]$. For each $k = 1, 2, \dots, n$,

$$\inf\{f(x) \mid a_{k-1} \le x \le a_k\} = 0 = m_k.$$

Thus, $L(f,P) = 0$. For the supremum,

$$\sup\{f(x) \mid a_{k-1} \le x \le a_k\} = 1 = M_k.$$

Thus, $U(f,P) = 1$. Then finally

$$L(f) = \sup_P\{L(f,P)\} = 0 < 1 = \inf_P\{U(f,P)\}.$$

So f is not Riemann integrable.

Remark 2.

(1) If $[a,b]$ is fixed, then

$$\underline{\int} f = L(f), \quad \overline{\int} f = U(f).$$

(2) The Dirichlet function is not Riemann integrable, however, it is Lebesgue integrable.

Criteria for Integrability

Theorem 7.2.8 (Integrability Criterion). *A bounded function f is integrable on $[a,b]$ if and only if, for every $\epsilon > 0$, there exists a partition P_ϵ of $[a,b]$ such that*

$$0 \le U(f,P_\epsilon) - L(f,P_\epsilon) < \epsilon.$$

Proof.

($\Rightarrow$)

Assume that f is integrable, i.e., $L(f) = U(f)$. Let $\epsilon > 0$. Because $U(f)$ is the greatest lower bound of the upper sums, we know that given some $\epsilon > 0$, there must exist a partition P_1 such that

$$U(f,P_1) < U(f) + \frac{\epsilon}{2}.$$

Likewise, there exists a partition P_2 satisfying

$$L(f,P_2) > L(f) - \frac{\epsilon}{2}.$$

Let $P_\epsilon = P_1 \cup P_2$ be the common refinement. Since f is integrable

$$\begin{aligned} U(f,P_\epsilon) - L(f,P_\epsilon) &\leq U(f,P_1) - L(f,P_2) \\ &< \left(U(f) + \frac{\epsilon}{2}\right) - \left(L(f) - \frac{\epsilon}{2}\right) \\ &= \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

($\Leftarrow$)

Let $\epsilon > 0$. If such a partition P_ϵ exists, then

$$U(f) - L(f) \leq U(f,P_\epsilon) - L(f,P_\epsilon) < \epsilon.$$

Since ϵ was chosen arbitrarily, we must have that $U(f) = L(f)$, so f is integrable. □

Let's use how the Integrability Criterion can be used to prove that a function is integrable.

Example 27. Let $f : [a,b] \to \mathbb{R}$ be constant. Show that f is integrable.

Proof. Let $\epsilon > 0$ and $f(x) = k$ for $x \in [a,b]$. We define P_ϵ to be the partition $\{a_0 = a, a_1 = b\}$. This partition has the upper sum

$$\begin{aligned} U(f,P_\epsilon) &= \sum_{k=1}^{n} M_k(a_k - a_{k-1}) \\ &= M_1(a_1 - a_0) \\ &= \sup\{f(x) \mid a_{k-1} \leq x \leq a_k\} \cdot (b-a) \\ &= k(b-a). \end{aligned}$$

Note that $\sup\{f(x) \mid a_{k-1} \leq x \leq a_l\}$ is k because f is constant. Similarly, we can compute the lower sum

$$\begin{aligned} L(f,P_\epsilon) &= \sum_{k=1}^{n} m_k(a_k - a_{k-1}) \\ &= \inf\{f(x) \mid a_{k-1} \leq x \leq a_k\} \cdot (b-a) \\ &= k(b-a). \end{aligned}$$

From this, we have that

$$0 \leq U(f,P_\epsilon) - L(f,P_\epsilon) = 0 < \epsilon.$$

Thus, a constant function $f : [a,b] \to \mathbb{R}$ is integrable. □

We can extend this to show that any continuous function is integrable.

Theorem 7.2.9. *If f is continuous on $[a,b]$, it is integrable.*

Proof. Since f is continuous on a compact set $[a,b]$, it is bounded. It is also uniformly continuous for the same reason. Let $\epsilon > 0$. There exists $\delta > 0$ such that whenever $|x - y| < \delta$, we

have

$$|f(x)-f(y)|<\frac{\epsilon}{b-a},$$

for $x,y\in[a,b]$. Choose $P_\epsilon=\{x_0=a,x_1,\dots,x_n=b\}$ such that $|x_{k_n}-x_{k-1}|<\delta<$ for $k=1,2,\dots,n$. Since f is continuous on $[x_{k-1},x_k]$ for $k=1,2,\dots,n$, by the Extreme Value Theorem, f has $y_k,z_k\in[x_{k-1},x_k]$ such that $f(y_k)=m_k$ and $f(z_k)=M_k$. Thus

$$\begin{aligned}L(f,P_\epsilon)&=\sum_{k=1}^{n}m_k\Delta x_k=\sum_{k=1}^{n}f(y_k)\Delta x_k\\ U(f,P_\epsilon)&=\sum_{k=1}^{n}M_k\Delta x_k=\sum_{k=1}^{n}f(z_k)\Delta x_k.\end{aligned}$$

We have that

$$\begin{aligned}0&\le U(f,P_\epsilon)-L(f,P_\epsilon)\\ &\le\sum_{k=1}^{n}(f(z_k)-f(y_k))\underbrace{\Delta x_k}_{<\delta}\\ &<\sum_{k=1}^{n}\frac{\epsilon}{b-a}\Delta_k\\ &=\frac{\epsilon}{b-a}\sum_{k=1}^{n}\Delta_k\\ &=\frac{\epsilon}{b-a}(b-a)\\ &=\epsilon.\end{aligned}$$

Thus, f is integrable. □

Remark 3. *$f:[a,b]\to\mathbb{R}$ can be piecewise continuous, in which case we just integrate on each subinterval.*

Integrating Functions with Discontinuities

Theorem 7.3.2. *If $f:[a,b]\to\mathbb{R}$ is bounded, and f is integrable on $[c,b]$ for all $c\in(a,b)$, then f is integrable on $[a,b]$. An analogous result holds at the other endpoint.*

Proof. Let $\epsilon>0$. We want to find P_ϵ such that

$$\begin{aligned}U(f,P_\epsilon)-L(f,P_\epsilon)&=\sum_{k=1}^{n}(M_k-m_k)\Delta x_k\\ &=(M_1-m_1)(c-a)+\sum_{k=2}^{n}(M_k-m_k)\Delta x_k\\ &<\epsilon.\end{aligned}$$

Since f is bounded, there is $M > 0$ such that $|f(x)| < M$ for $x \in [a,b]$. Choose $c \in (a,b)$ so that

$$c - a < \frac{\epsilon}{4M}.$$

Since $f : [c,b] \to \mathbb{R}$ is integrable, there exists a partition P_1 such that

$$\sum_{k=1}^{m} (M_k - m_k)\Delta x_k < \frac{\epsilon}{2},$$

where $P_1 = \{c = x_1, x_2, \ldots, x_n = b\}$. Define the partition $P_\epsilon = \{a\} \cup P_1$ on $[a,b]$. Thus

$$\begin{aligned}
0 &\leq U(f, P_\epsilon) - L(f, P_\epsilon) \\
&\leq \underbrace{(M_1 - m_1)}_{} < 2M \underbrace{\Delta x_1}_{<\epsilon/4M} + \underbrace{\sum_{k=2}^{n} (M_k - m_k)\Delta x_k}_{<\epsilon/2} \\
&= 2M \cdot \frac{\epsilon}{4M} + \frac{\epsilon}{2} \\
&= \frac{\epsilon}{2} + \frac{\epsilon}{2}.
\end{aligned}$$

Thus, for every $\epsilon > 0$, we can find P_ϵ such that the difference of the upper and lower sums are less than ϵ. □

Properties of the Integral

Theorem 7.4.1. *Assume $f : [a,b] \to \mathbb{R}$ is bounded, and let $c \in (a,b)$. Then, f is integrable on $[a,b]$ if and only if f is integrable on $[a,c]$ and $[c,b]$. In this case, we have*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof. ??? □

From Calculus I and II, we've seen algebra of integration. This makes a comeback in Advanced Calclulus.

Theorem 7.4.2. *Assume f and g are integrable functions on the interval $[a,b]$.*

(i) The function $f + g$ is integrable on $[a,b]$ with $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

(ii) For $k \in \mathbb{R}$, the function kf is integrable with $\int_a^b kf = k\int_a^b f$.

(iii) If $m \leq f(x) \leq M$ on $[a,b]$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$.

(iv) If $f(x) \leq g(x)$ on $[a,b]$, then $\int_a^b f \leq \int_a^b g$.

(v) The function $|f|$ is integrable and $\left|\int_a^b f\right| \leq \int_a^b |f|$.

Proof. To prove property (i), we have that since f and g are integrable, there are sequences of partitions P_n and Q_n satisfying

$$\lim_n [U(f, P_n) - L(f, P_n)] = 0,$$

and

$$\lim_n [U(g, Q_n) - L(g, Q_n)] = 0.$$

For each $n \in \mathbb{N}$, define $R_n = P_n \cup Q_n$ on $[a, b]$. Since R_n is a refinement of P_n and Q_n for $n \in \mathbb{N}$, we have

$$L(f, P_n) \leq L(f, R_n) \leq U(f, R_n) \leq U(f, P_n)$$

and also

$$L(g, Q_n) \leq L(g, R_n) \leq U(f, R_n) \leq U(g, Q_n).$$

Note that

$$L(f + g, R_n) = L(f, R_n) + L(g, R_n).$$

The same is true for the upper sum. Thus,

$$\begin{aligned} \lim_n [U(f + g, R_n) - L(f + g, R_n)] &= \lim_n U(f, R_n) + \lim_n U(g, R_n) - [\lim_n L(f, R_n) + \lim_n L(g, R_n)] \\ &= 0. \end{aligned}$$

□